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REVIEW

Survey on legal information extraction: current status and open challenges

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Abstract

The goal of information extraction is to extract structural knowledge (such as entities, relations and events) from plain and unstructured texts. Information extraction in legal documents has recently gained a lot of attention in the natural language processing (NLP) community due to the high demand for efficient information extraction for legal practitioners and companies. Given that the legal documents are unique and their processing is challenging, there is a pressing need for applications of NLP techniques to tackle these challenges. In this research, we present a survey on the recent advancements in legal information extraction focusing on three tasks: named entity recognition, relationship extraction and event detection. We report language resources and systems in multiple jurisdictions and languages for each task. Based on the thorough review conducted, we identify insights into the techniques employed and promising research directions that merit further exploration in future studies. We maintain a public repository and consistently update related resources at https://github.com/DamithDR/legalinformationextraction.

Keywords Legal information extraction · Legal NLP · Named entity recognition · Relationship extraction · Event detection

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1 Introduction

There is an increasing volume of publicly available data, and developing tools to identify important information from these data is an important task with many practical applications. However, much of this data is available in the form of unstructured or semi-structured texts [1]. Using humans to extract information manually from these texts will be costly and highly inefficient [2] given the large volume of data available. *Information Extraction* (IE) emerged as a solution to this problem. IE is a crucial application of natural language processing (NLP) as it transforms unstructured text into organised information in the form of structural knowledge [3] automatically. Furthermore, IE is essential as a fundamental requirement for various applications, including the construction of knowledge graphs [4, 5], knowledge reasoning [6, 7] and question answering [8, 9].

An IE process can be very complex. Therefore, it is common to decompose it into several tasks. Over the years, the following have established themselves as main tasks in an IE process.

1. **Named entity recognition** (NER) refers to extracting substrings within a text that name real-world objects and to determine their type (for example, whether they refer to persons or organisations) [10].
2. **Relationship extraction** (RE) occurs after NER and refers to classifying the relation type between two given entities [11].
3. **Event detection** refers to the task of identifying and extracting information about events from textual data. This involves recognising instances in the text where an event is described and extracting relevant details about it, such as the participants, time, location and the nature of the event itself [12].

The evolution of information extraction technology mirrors the progression of natural language processing. It has progressed through three stages: rule and dictionary-based methods, statistical machine learning approaches, and, more recently, deep learning methodologies [13]. Recent deep learning methods such as transformers [14] have provided state-of-the-art results in all the IE tasks mentioned above [15]. However, most of these IE methods are developed and tested in the general domain. The research advancements in the general domain have been discussed widely in recent surveys [3, 10, 11, 13].

NLP techniques, including the IE techniques developed for the general domain, might not work for a specific domain mainly because the structure and terms used in particular domains differ significantly from the general domain, and the information required to extract differs from domain to domain. Therefore, different IE methods have been developed for domains such as clinical [16] and biomedical [17]. These methods have been widely discussed in recent domain-specific IE surveys [18, 19].

1.1 Motivation for a legal IE survey

One such domain that differs significantly from the general domain is the legal domain. On the one hand, the legal domain stands out as a particularly compelling area of interest for information extraction. This is primarily due to the escalating volume of legal documents, such as court cases and contracts, which burdens legal professionals who must dedicate extensive time to reading and understanding these materials [20, 21]. Therefore, there is a growing interest within the research community in leveraging NLP techniques for information extraction in legal texts, making it a topic of increasing significance.

On the other hand, the legal domain has its own set of challenges. As aforementioned, legal language differs considerably from general language and has complex meanings for specific terms and phrases [22]. Therefore, IE approaches that are commonly effective in general text might perform poorly in legal texts, and models trained on general corpora may yield unsatisfactory results when applied to legal contexts [23]. Furthermore, most legal documents are very lengthy, and the information is scattered. This nature of the legal documents makes it harder for specific methods to identify and extract information from the given text correctly. For example, transformer-based pre-trained language models that provide state-of-the-art results in the general domain [24] often struggle in the legal domain as they cannot handle longer contexts [25, 26]. However, the significance of legal information in decision-making cannot be overstated, as inaccurate or omitted information may lead to undesirable consequences. Therefore, ensuring high accuracy in information extraction systems within the legal domain is of paramount importance.

The last few years saw a rapid growth in NLP techniques applied to the legal domain broadly referred to as *LegalNLP* [27]. Natural Legal Language Processing (NLLP) [28–30],[1] Mining and Learning in the Legal Domain (MLLD) [31][2] workshops organised with major computational linguistics conferences, shared tasks organised such as LegalEval [32] and AILA (Artificial Intelligence for Legal Assistance) [33] have fuelled LegalNLP. The recent surveys [34–38] extensively discussed the LegalNLP techniques. However, these surveys have mainly discussed areas such as legal judgment prediction [39, 40], legal text summarisation [41, 42], legal question answering [43, 44], legal information retrieval[3] [46, 47] and legal text representation [48, 49] and do not focus on legal information extraction. Understanding the importance of documenting available language resources and computational methods, this paper presents a survey on legal information extraction.

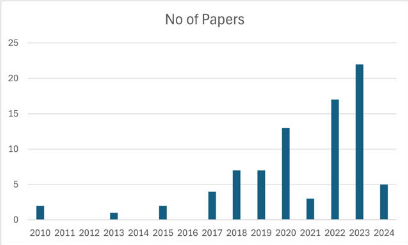
Solihin et al. [50] presented a survey on advancements in information extraction use in legal documents. The survey only includes research articles published before 2020 December. However, there was a major paradigm shift in legal NLP after 2020, with the introduction of pre-trained transformer models such as `LEGAL-BERT` [25], which has provided state-of-the-art results in many legal NLP tasks. However, IE research based on pre-trained transformer models was not included in the previous survey. Furthermore, while Solihin et al. [50] present a detailed overview of the research papers related to legal IE, they do not provide details of the individual papers. Therefore, crucial information such as dataset availability and annotated named entities remains undisclosed. To address these shortcomings, our survey includes the latest research in legal IE and provides a thorough examination of each paper, filling the gap in knowledge left by the previous survey.

Historically, IE tasks such as NER have been considered important in evaluating language understanding of machine learning models. For example, multilingual adaptations of the General Language Understanding Evaluation benchmark (GLUE) [51], such as bgGLUE (Bulgarian) [52], Indic-GLUE (Indian languages) [53], Sin-GLUE [54] and KLUE (Korean) [55] encompass NER as one of the tasks. Interestingly, while they include IE tasks in their benchmarks, there exists a notable absence of IE tasks within legal natural language processing benchmarks. Notably, LexGLUE [56], a benchmark tailored for legal language understanding, lacks any tasks related to legal IE, which we believe is a significant limi-

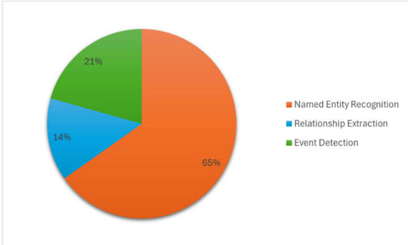
[1] Information about NLLP is available at https://nllpw.org/.

[2] Information about 2023 workshop is available at https://sites.google.com/view/mlld2023/.

[3] Information retrieval focuses on retrieving relevant documents or information in response to user queries while information extraction focuses on automatically extracting structured information from unstructured text data [45].



(a) Number of papers published over time on legal Information Extraction



(b) Composition of papers in different legal information extraction tasks: named entity recognition, relationship extraction and event detection.

Fig. 1 Overview of legal information extraction literature: **a** publication trends over time and **b** task-wise distribution

tation. We hope that the insights and advancements presented in this survey will pave the way to include legal IE tasks in future legal NLP benchmarks.

1.2 Building a collection of legal IE papers

In line with previous IE surveys conducted on the general domain [13, 57], this survey focuses on three main tasks within the IE process that we discussed before: (*Task A:*) named entity recognition, (*Task B:*) relationship extraction and (*Task C:*) event detection.

We constructed a collection of nearly all published Legal IE papers, taking the past decade as our window of analysis. To build this collection, we started by reviewing the proceedings of general NLP conferences (ACL, NAACL, EMNLP, EACL, LREC, COLING and RANLP) through the ACL Anthology[4] database, several information retrieval/ extraction conferences (SIGIR, ECIR and FIRE), several knowledge discovery and databases conferences (PKDD and CIKM), several machine learning conferences (NeurIPS, ICML and ECAI) and several speciality Legal NLP gatherings (NLLP [28–30], MLLD [31], Jurix [58, 59] and ICAIL [60, 61]). Next, we performed iterative queries using keywords such as *Legal IE, Legal NER, Legal Relationship Extraction, Legal RE and Legal Event Detection* on major journal databases such as Springer, Elsevier, IEEE, Cambridge University Press, Oxford University Press, Frontiers, and MDPI. This initial screening yielded a large number of results, and we then performed manual filtering only to keep relevant articles to legal IE. Barring exceptional situations, we restricted our corpus to include only peer-reviewed scientific journals and technical conference proceedings. We excluded pre-prints, essays or blog posts from our survey.

This process finally resulted in 81 papers. We believe we have acquired nearly every paper published on legal information extraction over the past decade. Figure 1a provides a time-series representation of the volume of papers published over the past decade on this topic. We observe a significant increase in the total yearly volume of publications.

As the last step, we manually categorise each paper into one or more tasks that we defined before as in Fig. 2. The distribution of papers across these tasks is visually depicted in Fig. 1b, showcasing the percentage representation of each task within our paper corpus. As shown there, more than half of the articles in our corpus discuss legal named entity recognition.

[4] Available at https://aclanthology.org/.

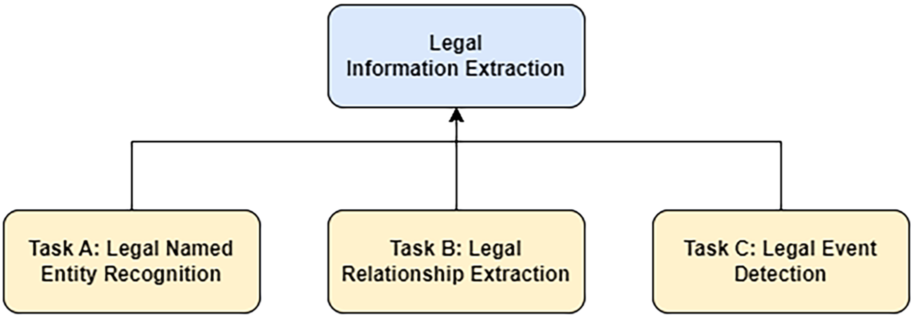


Fig. 2 Overview of the survey

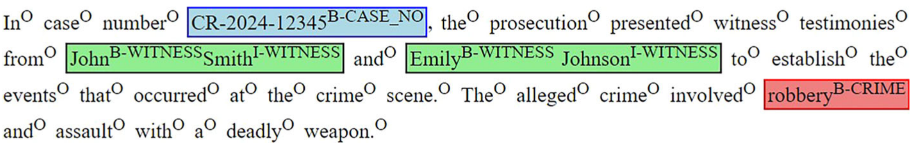


Fig. 3 Example sentence with BIO-tagged legal named entities

However, there are still a significant number of papers for legal relationship extraction and legal event detection. In the following sections, we describe these papers. The rest of the paper is structured as follows. Section 2 discusses the research related to legal named entity recognition, Sect. 3 describes research related to legal relationship extraction and Sect. 4 discusses research related to legal event detection. All three sections discuss available language resources and computational methods separately. We finish the paper with conclusions, future directions and open challenges in legal information extraction.

2 *Task A:* legal named entity recognition

Named entities are specific words or phrases that refer to real-world objects such as people, organisations, locations and dates. NER aims to locate and categorise these entities into pre-defined categories [10, 62, 63]. NER is a crucial component of an IE process, which aims to identify and classify named entities within the text. As we will describe in the subsequent sections, NER is a predecessor for relationship extraction and event detection [64]. Recent surveys such as Jehangir et al. [65], Guellil et al. [66] and Li et al. [67] provide a comprehensive view of NER datasets and methods in the general domain. In the general domain, there are common pre-defined types of entities, such as person names, locations, organisations and dates. However, these general entity types do not hold much significance within the legal domain. Therefore, a specific set of named entities is essential in the legal field instead of the general entity types. Since these entities only exist in the legal documents, the general systems do not recognise them and need further research on datasets (Sect. 2.1) and methods (Sect. 2.2) to capture legal entities.

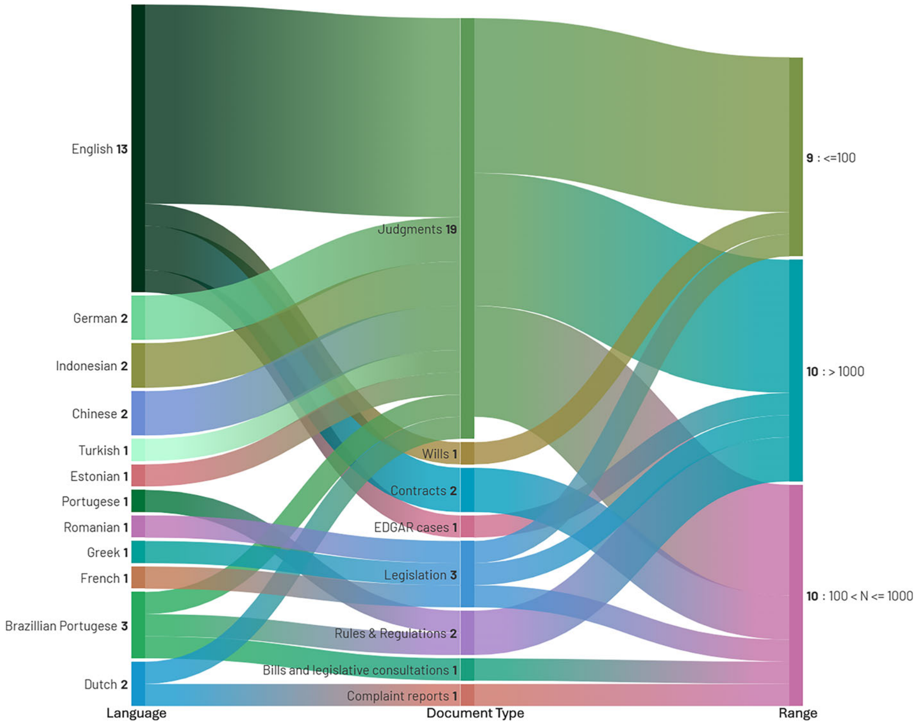


Fig. 4 Analysis of NER datasets by their Language, Domain and size. N—number of documents used

2.1 Legal NER data

NER has been commonly modelled as a supervised machine learning task in which the machine learning models are trained on a task-specific annotated dataset. Therefore, adequately annotated datasets are a major requirement in NER tasks. Furthermore, recent advancements in NLP with deep learning models require a considerable amount of data to fine-tune their weights. Over the years, researchers have created different legal NER datasets in different jurisdictions and languages. In this section, we will explore NER datasets in detail.

Getting inspired from general NER datasets such as CoNLL-2003 [68], all the legal NER datasets followed the BIO tagging system, where BIO refers to B—Beginning, I—Inside and O—Outside a named entity. Figure 3 shows an example tagging for a sentence where B-CASE_NO is used to indicate the beginning of a court case number, B-WITNESS for the beginning of the name of a witness, I-WITNESS for inside the name of a witness, B-CRIME for the beginning of a crime event and O elsewhere.

Figure 4 shows an analysis of the datasets used for NER. English overwhelmingly dominates the landscape, contributing datasets to all domains and spanning the full range of dataset sizes, with a notable portion exceeding 1000 records. In contrast, other languages such as German, Indonesian, Chinese and Dutch are limited to fewer domains—primarily judgments and contracts and are generally associated with smaller dataset sizes, often less than 100. Among the domains, judgments are the most frequently represented, drawing from a wide variety of languages, while other types, such as wills, complaint reports, legislation,

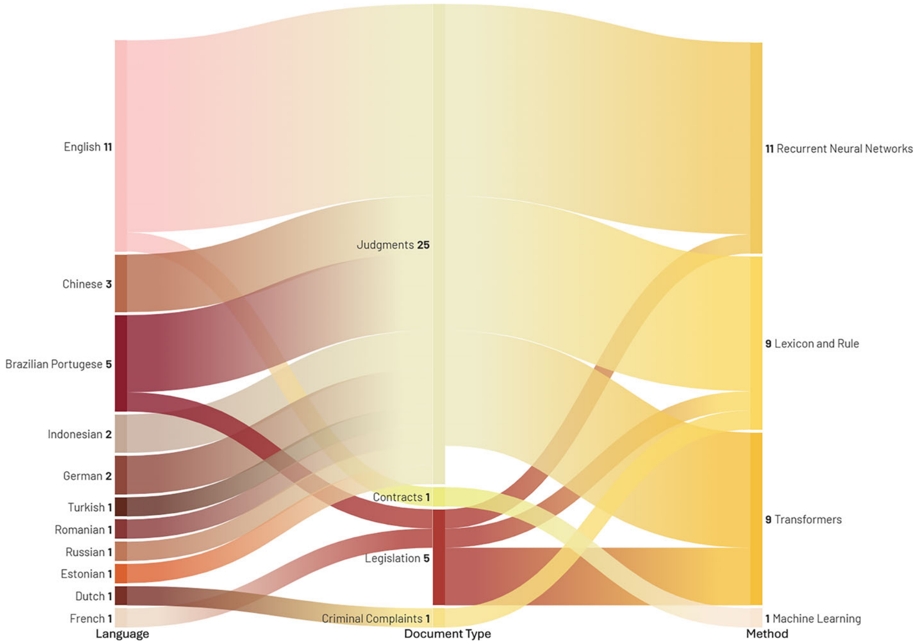


Fig. 5 Analysis of NER methods by their language, domain and method

and rules and regulations, are more niche and typically supported only by English-language datasets. Most non-English datasets fall into the smallest size bracket (≤ 100), indicating limited data availability for these languages. This imbalance reveals a significant bias towards English and highlights the underrepresentation of both non-English languages and diverse legal document types.

The datasets are summarised in Table 1. A complete, detailed list of datasets can be found in "Appendix A".

2.2 Legal NER methods

As shown in the previous section, legal NER datasets have been constructed on different types of legal documents, such as court cases, legal contracts and code of conduct. These different legal document types have different characteristics with regard to length and word frequency, which imposes the requirement of studying different models on different data in the legal domain.

NER methods can be categorised into four main approaches: (1) Lexicon-based or rule-based, (2) recurrent neural network (RNN)-based, (3) machine learning-based and (4) transformer-based. Early approaches in legal NER have mainly used lexicon-based or rule-based methods. However, recent developments in large pre-trained language models have provided state-of-the-art results.

Figure 5 shows an analysis of the methods employed for NER task. The vast majority of the methods have focused on NER in judgment texts. A greater portion of English and Brazilian Portuguese work is on Judgments, while work on Chinese, Indonesian and German has entirely focused on judgments. Even though recurrent neural networks are the most

Table 1 Legal NER datasets

Name/reference	Named entities	Language	Jurisdiction	Availability
Ahmed et al. [21]	Per, Loc, Org, CaseNo, Resp, Date, Refcourt, Refcase, Ref, Appealcourt, Appealcaseno, Money, FIRno, Approved	English	Pakistan	No
Duarte et al. [69]	LREF, TREF, NE_ADM TIME_DURATION, TIME_DATE_REL_TEXT	Portuguese	Portugal	No
Luz de Araujo et al. [70]	ORGANIZACAO, PESSOA, TEMPO, LOCAL JURISPRUDENCIA, LEGISLACAO	Brazilian Portuguese	Brazil	No
Leitner et al. [71]	PER, LOC, ORG, NRM, REG, RS, LIT (19 Fine grained classes)	German	German	Yes
Cardellino et al. [23]	PERSON, ORGANISATION, DOCUMENT, ABSTRACTION, ACT	English	Europe	No
Au et al. [72]	LOCATION, PERSON, BUSINESS, GOVERNMENT, COURT, LEGISLATION/ACT, MISCELLANEOUS	English	United States	Yes
Çetindağ et al. [73]	PER, LOC, ORG, DAT, LEG, COU, REF, OFF, O	Turkish	Turkey	Yes

Table 1 continued

Name/reference	Named entities	Language	Jurisdiction	Availability
Orasmaa et al. [74]	PER, LOC-ORG, LOC, ORG, MISC	Estonian	Estonia	Yes
Brugman [75]	CASE, LOC, LEG, ID, DATE, SECTION, NJ, ECLI, PRISON_SENTENCE, FINANCIAL_SENTENCE, DURATION, CURRENCY, PROBATION, CONVERSION	Dutch	Netherland	No
Kamat et al. [76]	CRT, PT, CD, DOC, JURD, LOC, CTYP, AUTH, CRTOF, DOJUD (more finegrained classes)	English	India	Yes
Kalamkar et al. [77]	COURT, PETITIONER, RESPONDENT, JUDGE, LAWYER, DATE, ORG, GPE, STATUTE, PROVISION, PRECEDENT, CASE_NUMBER, WITNESS, OTHER_PERSON	English	India	Yes
Naik et al. [78]	PERSON, LOC, DATE, ORG, COURT, LEGAL, ACT, CASE_NO	English	India	No
Nuranti and Yulianti [79]	ADVOKAT, AMAR, HAKIM, JAKSA ORGANISASI, PANITERA, PERATURAN, PUTUSAN TANGGAL, TERLIBAT	Indonesian	Indonesia	No
Pais et al. [80]	Person, LOCATION, ORGANISATION, TIME, LEGAL REF	Romanian	Romania	Yes
Andrew [81]	PERSONNE, NOM, ADDRESS, SOCIETE_PRINCIPALE, SOCIETE_SECONDAIRE, ROLE, FONCTION, TYPE_SOCIETE	French	Luxembourg	No

Table 1 continued

Name/reference	Named entities	Language	Jurisdiction	Availability
Glaser et al. [82]	PER, ORG, LOC, DA, MV, REF, OTH	German	Germany	No
Krasadakis et al. [83]	LAW, CASE LAW, EURO LAW, EURO CASE LAW, ADMIN DOC, DOCTRINE, BOOK, FOREIGN LAW	Greek	Greece	No
Jayasooriya et al. [84]	case titles, acts, legal citations, counsel, judges, argued dates, verdict dates, judgments	English	Sri Lanka	No
Shi et al. [20]	N, Np, M, Nm, S, D, P, A, O	Chinese	China	No
de Almeida et al. [85]	P, O	English	United States	Yes
Samarawickrama et al. [86]	P, O	English	United States	No
Solihin and Budi [87]	Nomor Putusan, Nama terdakwa, Tindak pidana Melanggar KUHP, Tuntutan hukuman, Putusan hukuman Tanggal putusan, Hakim Ketua, Hakim anggota Panitera, Penuntut umum	Indunesian	Indonesia	No
Chen et al. [88]	Person, Location, Organisation	Chinese	China	No
Sharafat et al. [89]	CaseNo, Date, Loc, Money, Org, Per, Ref, RefCase, RefCourt, Resp	English	Pakistan	No

Table 1 continued

Name/reference	Named entities	Language	Jurisdiction	Availability
Chalkidis et al. [90]	Contract Title, Contracting Parties, Start Date, Effective Date, Termination Date, Contract Period, Contract Value, Governing Law, Jurisdiction, Legislation Refs, Clause Headings	English	Not Mentioned	Yes
Schraagen et al. [91]	location, person, organisation, event, product, miscellaneous	Dutch	Netherlands	No
Albuquerque et al. [92]	DATA, EVENTO, FUNDlei, FUNDapelido, FUNDprojetodelei, FUNDsolicitacaotrabalho, LOCALconcreto, LOCALvirtual, ORGpartido, ORGgovernamental, ORGnaogovernamental, PESSOAindividual, PESSOAgrupoind, PESSOAcargo, PESSOAgrupocargo, PRODUTOsistema, PRODUTOprograma, PRODUTOoutros	Brazilian Portuguese	Brazil	Yes
Kwak et al. [93]	Testator, Trigger, Condition, Beneficiary, Will, Asset, Executor, Witness, Time, Duty, State, Date, County, Right, Expense, Debt, Bond, Codicil, Trustee, Non-beneficiary, Tax, Affidavit, Notary public, Trust, Guardian, Conservator	English	United States	Yes

Table 1 continued

Name/reference	Named entities	Language	Jurisdiction	Availability
Oliveira et al. [94]	contract_number, GDF_process, contractual_parties, contract_object, contract_date, contract_value, contract_duration, budget_unit, work_program, nature_of_expenditure, commitment_note	Brazilian Portuguese	Brazil	No
Aejas et al. [95]	Title, Parties, EffectiveDate, TermLength, NotificationPeriod, GoverningLaw	English	United States	Yes

Column *Name/Reference* refers to the reference of the paper that introduced the dataset. Column *Named entities* refers to the annotated/considered named entities in the dataset, column *Language* shows the language used in the documents in the dataset, and column *Jurisdiction* refers to the jurisdiction to which the documents in the dataset belong. The column *Availability* expresses whether the dataset is publicly available. If the dataset is publicly available, we have provided the link to the dataset as a hyperlink used architecture of the methods, transformers and lexicon-based methods have also been used in similar amounts. However, it is notable that the NER methodologies have not been experimented with equally in different legal documents. While judgment texts have received much attention from the research community, other document types like contracts still remain understudied.

2.2.1 Lexicon- and rule-based legal NER methods

Rule-based methods automate the extraction of named entities based on a set of linguistic rules. These methods effectively generate rules by utilising different resources [96, 97].

Andrew [81] combined regular expressions with conditional random fields (CRFs) [98] to extract named entities. CRFs were employed to capture coarse-grained entities, while rules were utilised for finer-grained entity identification. The authors presented results for both exact and partial matching, conducting experiments on a French dataset detailed in the data Sect. [81]. Their findings indicate precision exceeding 95%, but recall stands at a modest 47.93%. The authors noted that the rules applied in the training set were directly transferred to the test set, suggesting the need for adjustments to better align with the test set and enhance results.

Glaser et al. [82] demonstrate the effectiveness of rule-based methodologies through their experiments comparing three distinct approaches: (1) GermaNER, (2) DBpedia Spotlight and (3) Templated NER, using the dataset discussed in the data section. GermaNER and DBpedia are integrated with the Apache UIMA[5] pipeline. The authors highlight the practical challenges associated with deploying their Template-based method on judgments, restricting its use only to contract data. The Template-based approach is characterised by its simplicity and directness, wherein a contract template containing key elements is maintained, leaving spaces for variable content (e.g. Name:...). Subsequently, the authors apply this template to actual contracts, computing the difference between the contract and the template. Employing this methodology, they achieve an F1 score of 0.92. However, the authors note the inability to evaluate this method on court cases due to the lack of conformity in court proceedings with the template structure.

Solihin and Budi [87] experimented with NER in court cases using three folded NLP pipelines. First, they extracted the relevant fragments from the case document using keywords matching. They crafted different terms and keywords for each fragment such that the appearance of these in the text was extracted as fragments. Then, the fragments were tokenised into words, punctuation, etc. Finally, the tokens were carefully examined, and they created rules according to the named entities such that the regular expression pattern matcher could identify the named entities from the tokenised sentence. They could achieve a 0.89 F1 score with this method, suggesting the rules could be employed to a uniform structured documents for the NER task.

Kadu and Ashwini [99] used an open information extraction method to extract the named entities in Indian court cases. Their entity extraction method is based on part-of-speech (POS) tagging. After the POS tagging of the cleaned information, they follow a method called named entity chunking. In other words, they find the related nouns around a verb with the help of prepositions. However, the authors have not presented empirical results in their paper.

Thomas and Sangeetha [100] employed rules as one of their three approaches to the NER task in Indian case judgements. They crafted regular expressions based on trigger words to detect Person, Location and Organisation. However, trigger words like Mr, Ltd, Inc are not

[5] Available at https://uima.apache.org/.

always present in the texts. As a matter of fact, they used a DBpedia as a lookup table to match the entities. Yet they highlight the drawbacks of this approach as even DBpedia does not contain all the names of the locations and organisations.

Schraagen et al. [91] experimented NER using Dutch criminal complaints we mentioned in the data section. They used Frog NER[6] as their NER system and compared the results with human annotators. They report a 0.71 F1 score for the named entity detection task.

Niu and Zeng [101] performed a joint method for litigants extraction (JMLE) in Chinese judgements. Their approach is based on linguistic structure in judgements, where if a litigant is mentioned in the text, it follows: indicators (plaintiff, defendant, legal representative, etc.) + entity object + other contents. The authors utilise this structure to extract litigant entities with a three-staged NLP pipeline, namely text processing, candidate entities generation and litigant entities generation. In the first two stages, they clean the text and find the candidate short texts which contain litigant entities, and in the last stage, they utilise a comprehensive dictionary to filter the litigant entities. Their results report that the JMLE method could achieve a 0.8937 F1 score, surpassing the baselines they considered in their experiment.

Sharafat et al. [89] performed NER on Pakistani judgements using three models: hidden Markov model (HMM), maximum entropy Markov model (MEMM) and conditional random fields (CRFs). They conducted twelve different experiments based on the annotation and tagging settings. They followed two schemes of annotations (A1, A2), A2 being high-level and A1 in a more granular level. They used two tagging mechanisms, namely IO and IOB schemes. They report CRF as their best performing model, surpassing both HMM and MEMM models in all settings. They also point out the effect of different annotation schemes, where A2 shows a 0.17 F1 score improvement in the IOB tagging scheme and 0.25 in the IO tagging scheme.

Dozier et al. [22] reported experimental research on three main approaches: (1) namely lookup, (2) context rules and (3) a statistical model on US legal texts. The Namely lookup leverages a pre-defined set of words (word list), and in the process of NER, the system will match these words with the legal text. This is very simple but effective in unambiguous entities such as drug names. Often, these names are unique and unusual, which makes the lookup method perform better in them. However, the downside of this approach is that if there are ambiguous data, such as person names, this method could produce false positives. However, the methods are not evaluated in any of the datasets.

The rule-based methods exhibit notable effectiveness, particularly when the documents follow a uniform structure [82]. However, this uniformity is not common in many legal documents. In the context of court cases, for instance, the structure could change from one court to another, even in the same jurisdiction. Consequently, creating generalised rules for extracting information from legal documents becomes exceedingly challenging due to this lack of uniformity.

2.2.2 Machine learning-based legal NER methods

Chalkidis et al. [90] extracted 11 different types of contract elements using a sliding window approach with logistic regression (LR) [102] and support vector machines (SVM) [103]. In both approaches (sw-lr-emb, sw-svm-emb), they used one classifier for each contract element type resulting 11 classifiers per experiment. The task of the classifier is to binary output whether the input phrase contains the relevant contract element type. The phrasing was done using sliding window approach 5–6 tokens. Finally, the window becomes a concatenated

[6] Available at https://languagemachines.github.io/frog/.

feature vector, which is the input to the classifier. Their third and forth models have hand-crafted features instead of token embedding and they used same LR and SVMs as classifiers (sw-lr-hcf, sw-svm-hcf). Finally, they used a combination of above approaches (sw-lr-all), concatenated word embeddings and hand-crafted embeddings. Their results show that both classifiers perform best in the combined-all approach (sw-lr-all and sw-svm-all). Four out of eleven contract elements have same results for both sw-lr-all and sw-svm-all approaches: sw-svm-all classifier outperforms in seven contract element types and rest by sw-lr-all. Effective date is the only contract element type in which the sw-svm-hcf model performed best reporting 0.8 F1 score.

2.2.3 Recurrent neural networks-based legal NER methods

Long short-term memory networks (LSTMs) can be considered the most popular type of RNNs used for NER tasks. Conditional random fields (CRFs) [104] connected with LSTMs provide a powerful way of making named entity predictions and have been employed largely in the literature before the emergence of pre-trained transformer models even within the general domain [105].

Leitner et al. [106] offer an extensive analysis with a comparison of CRF-based models and BiLSTM-based models by utilising the dataset described in Leitner et al. [71] which we discussed before. Their analysis has different combinations of LSTM, CRF and CNNs. CRF experiment group consists of (1) CRF-F with features, (2) CRF-FG with features and gazetteers and (3) CRF-FGL with features, gazetteers, and a lookup table, while for BiLSTM, (1) BiLSTM-CRF (2) BiLSTM-CRF+ with character embeddings from the BiLSTM; (3) BiLSTM-CNN-CRF with character embeddings from CNN. As expected, BiLSTM model combinations outperform the CRF models. The F1 score for the fine-grained classification reaches 0.9546 and 0.9595 for the coarse-grained one. CRFs reach up to 0.9323 F1 for the fine-grained classes and 0.9322 F1 for the coarse-grained ones. Both models perform best for judge, court and law tags.

BiLSTMs and LSTMs have also been utilised by Luz de Araujo et al. [79], Çetindağ et al. [70], Nuranti and Yulianti [73], Pais et al. [80], Khasianov et al. [107] for NER in Indonesian, Brazilian, Turkish, Romanian and Russian legal documents. Nuranti and Yulianti [79] reported the best results using BiLSTM-CRF combination of 0.83 F1 score. They show that other neural models (LSTM-CRF, CNN) lag behind 2–12% in comparison with the BiLSTM-CRF model.

Luz de Araujo et al. [70] showed that LSTM-CRF architecture can perform better compared to ParamopamaWNN [108] in different Brazilian Portuguese NER datasets, HAREM [109] and WikiNER [110]. Çetindağ et al. [73] have experimented with Turkish legal NER with multiple word representation models, namely GloVe [111] and Morph2Vec [112], and they show the hybrid model which leverages both word representations and character-level features extracted with BiLSTM performs the best.

Pais et al. [80] introduced a baseline system on their Romanian national legislation data using a model architecture of BiLSTM+CRF. They have shown the results of each NER tag with combinations of different embedding representations. They present that their architecture could achieve over 0.75 F1 score on all tags. This illustrates BiLSTM+CRF combinations act as very good baseline systems in Legal NER.

de Almeida et al. [85] employ a sequence-to-sequence approach in their NE detection where they annotate the sentence with a 0, 1 sequence such that each word is annotated whether they are related to a legal party. Their system output is also a sequence of 1, 0, which was used to evaluate the results. Their approach suggests that the mask they introduced helped model to identify the relevant entities and through their altercation studies they report that among 64-, 128- and 256-layered architectures, 128 gives the best F1 score which is 0.56.

Khasianov et al. [107] introduced an intellectual system called "Robot Lawyer" having the main focus of dispute resolution in the entrepreneurship field. The system utilises the court proceedings to extract information using NER in Russian. They employed BiLSTM in combination with CRF to classify the entities. Their architecture uses a single-layer CNN network to create character-level embedding, and the input to the NER classifier is the concatenation of word embeddings and character-level embeddings. They have not published the performance of their system.

Wang et al. [113] used a combination of BERT, BiLSTM, Iterated Dilated Convolutional Neural Networks (ID-CNN) [114] and CRF models for the task of NER in Brazilian legal documents using LeNER-Br dataset described in the above section. `ID-CNN` is a method introduced as an alternative to BiLSTMs to reduce the time complexity in NER tasks. They used BERT to get contextual embeddings and then fed them to the `BiLSTM-CRF` network and `ID-CNN-CRF` network, resulting in two architectures. They report `BERT-ID-CNN-CRF` has a time gain of 50% against `BERT-BiLSTM-CRF` model. The third architecture is weight fusion of BERT to gain a single vector representation for the sequence. Finally, they used voting to fuse the heterogeneous models to obtain the best model Sequence Tagging Model (STM). Among all the model architectures, they report STM performs best with 0.9325 F1 score.

Chen et al. [88] have also used a BiLSTM architecture in combination with CRF to extract NEs in Chinese legal text, as mentioned in the data section. They utilise character-level embeddings where they feed them to a BiLTSM for the training process, followed by a CRF layer, which finally classifies the NER tags. Results show that their architecture could achieve 0.8015 F1 score, while the other approaches LSTM, BiLSTM, LSTM-CRF report, 0.6419, 0.7823, 0.6712 F1 scores, respectively.

Thomas and Sangeetha [100] used a combined architecture of BiLSTM, CRF and CNN (`BI-LSTM-CNN-CRF`). They used CoNLL dataset as their training data and Indian judgement cases as the test data. However, they explored this approach in combination with a knowledge base (KB), and they show the hybrid model of KB and deep learning (DL) model shows the best performance for entities; Location and Organisation reporting 0.94 and 0.71 F1 scores while this approach scores second best in Person entity marking 0.930 F1 score, while their KB + clustering approach gives 0.934 F1 score.

Krasadakis et al. [83] leveraged the CNNs provided by SpaCy.[7] They hypothesise that CNN models perform better than pre-trained transformer models due to the lack of resources for training a transformer model to perform NER in Greek legislation. Leveraging the Greek dataset mentioned before [83], they adopt an incremental training approach. After annotating a new batch of 10 documents, they re-train the model from scratch to assess performance improvements. They report that this method is useful to avoid model over-fitting and catastrophic forgetting. This model has predicted correctly around 89% entities in their test dataset.

Jayasooriya et al. [84] trained a spacy model on NER in Sri Lankan Supreme Court Verdicts. They used the spacy `en_core_web_lg` model for the named entity recognition task as the base model. The authors further fine-tuned the model on the Sri Lankan court data. Interestingly, they show over 90% accuracy on their test set [84].

Cardellino et al. [23] performed an NER at course-grained and fine-grained NEs. Their results show that simple single layer neural networks without word embeddings outperform support vector machines, Stanford NER [115] and neural networks with word embedding

[7] Available at https://spacy.io/.

methods. Further, they experimented with multiple neural network architectures having a different number of hidden layers, and they reported that a single hidden layer smaller than the input layer performed the best. Even though Stanford NER is compatible with fewer NER classes, the neural network delivers the best results with a higher number of NE types.

2.2.4 Transformers-based legal NER methods

Pre-trained transformer models [116] such as BERT [24] have delivered state-of-the-art results in many NLP tasks, including NER [117] in the general domain. Pre-trained models are very effective as they have a contextual understanding and are knowledgeable of different contexts. Furthermore, the development of domain-specific pre-trained transformer models such as `LEGAL-BERT` [25] has motivated researchers to use transformer models in legal NER tasks too.

Ahmed et al. [21] provide an empirical study on the two vanilla BERT [24] variations(`BERT-base-cased`, `BERT-base-uncased`) and the `LEGAL-BERT` [25] on the extraction of 14 named entities in Pakistani court cases [21], which we described in the previous subsection. The results show that the `BERT-base-cased` model performs the best in their case. Interestingly, the `BERT-base-cased` model has outperformed the `LEGAL-BERT` model, which is specifically trained on legal corpora and the `BERT-base-uncased` model. Since `LEGAL-BERT` is also trained on uncased corpora, they draw an assumption that a cased version of a `LEGAL-BERT` will produce better results.

Shi et al. [20] experimented with different combinations of `BiLSTM` and several pre-trained models. `RoBERTa-biLSTM-CRF` is formed as follows: RoBERTa model is used to replace the traditional word2vec to generate a word vector, and the text is transformed into a word vector as the input vector of `BiLSTM`. Then, `BiLSTM` extracts the feature of the input vector, obtains the semantic information of the context, and transmits the result of `BiLSTM` sent to CRF for text sequence annotation. Similarly, they experimented with `Lattice-CRF` inspired by `Lattice-LTSM` [118], which encodes a sequence of input characters as well as all potential words that match a lexicon, `BiLSTM-CRF`, `IDCNN-CRF` and `BERT-BiLSTM-CRF` too. They show that `RoBERTa-BiLSTM-CRF` outperforms all other model architectures with an F1 score of 0.8462 on their own "motor vehicle traffic accident liability dispute" data, to which we made a reference before.

While legal documents are written in English in most countries, some of them still use their native languages. This imposes the challenge of using the existing domain-specific pre-trained language models in the legal domain, such as `LEGAL-BERT` [25], as they are only available in English. Furthermore, training language-specific pre-trained models in the legal domain can be challenging due to the data availability. Therefore, researchers experimented with the performance of general pre-trained models in legal NER tasks. Duarte et al. [69] used `BERTimbau` [119] - pre-trained BERT models for Brazilian Portuguese in combination with `BiLSTM` and feed forward network (FFN) in their experiments in Portuguese NER. The transformer was used as the embedding layer, and they achieved an F1 score of 0.8867 in the NER task using their own dataset on Portuguese Consumer Law, which was described in the data section.

A recent study [74] experimented with monolingual pre-trained transformer models for Estonian such as `EstBERT` [120] and `Est-RoBERTa` on detecting named entities in nineteenth-century parish court records [74]. Despite the fact that these transformers were trained on a corpus with recent resources, the best model, `Est-RoBERTa`, recorded a micro-average F1 score of 0.936 in the NER task on these historic documents. Another positive outcome of this work is that even though `Est-RoBERTa` was pre-trained in the general domain, it produced very good results in the legal domain as well. This provides motivation to utilise general transformer models for legal information extraction tasks when domain-specific models are not available.

Transformer-based language models faced rapid developments in recent years and introduced LLMs, which follow the generative architecture. LLMS are pre-trained on massive corpora, with their primary objective being to predict the next word of the sequence. Recent studies have shown LLMs are capable of responding to conversations in a human-like manner and are able to continue to perform the NLP tasks competitively on request. Oleques Nunes et al. [121] studied evaluating Sabia [122], an LLM for Portuguese based on LLama2 [123] architecture. The authors used the existing datasets LeNER-Br and UlyssesNER-Br for Brazilian Portuguese, which we described in the data section. They evaluate how in-context learning strategies impact the final performance. Their heuristics are using (i) the top K similar examples, (ii) the top K similar examples chosen by entity type, and (iii) K random samples, as the examples in the prompt. They show that random sampling is the worst performing strategy for both datasets, and the Lener-Br dataset achieved a 0.51 F1 score with 8 samples, while UlyssesNERBr could achieve only a 0.48 F1 score with the same number of samples. The above-reported results were obtained in the top K similar examples setting, and they further report that changing the order of samples does not have much impact. This suggests the in-context learning strategy is sensitive to the samples provided in the prompt and needs to be carefully selected for better performance in NER systems using LLMs.

Oliveira et al. [94] studied incorporating LLMs (GPT3) as an annotation model to analyse the performance in NER in Brazilian Portuguese. They explored three different methods for annotations, namely LLM-based, weakly supervised (regular expressions and keywords) and human annotations. They evaluated three annotations(datasets) using Lener-BR,[8] `BERTimbau`,[9] `RoBERTa`[10] and `DistilBERT-PT`[11] models, which are specifically trained for Portuguese, except for RoBERTa. They report that even though the human annotations perform best in all cases, automated annotation approaches can produce competitive results. Their final results demonstrate preservation of human-trained models' scores, averaging 0.74 for GPT-3, 0.96 for weak supervision, 0.91 for GPT + weak supervision combination, and 0.84 for GPT + 30% human-labelling combination.

2.2.5 Hybrid legal NER methods

While the above sections present the work done using the different methods, the following approaches employ hybrid models. Combining `CRF` with neural models has shown success in many studies. Leitner et al. [106] combine `BiLSTM` with `CRF` models which shows superior performance to `CRF` only method. Luz de Araujo et al. [70] combined an `LSTM` with `CRF` model which outperformed ParamopamaWNN [108] in multiple Brazilian-portugese NER datasets. Similarly, [73] and [80] reported that the performance of joint `BiLSTM+CRF` models is superior compared to the `BiLSTM` models in Turkish and Romanian, respectively.

Khasianov et al. [107] further expand their experiments with a `CNN` layer in the model architecture as a combination of `BiLSTM`, `CNN` and `CRF`, which shows better performance over other methods. Wang et al. [113] put one more step forward by combining `BERT`,

8 Available at https://huggingface.co/pierreguillou/bert-base-cased-pt-lenerbr.

9 Available at https://huggingface.co/neuralmind/bert-base-portuguese-cased.

10 Available at https://huggingface.co/FacebookAI/roberta-base.

11 Available at https://huggingface.co/adalbertojunior/distilbert-portuguese-cased.

BiLSTM, CNN and CRF models, which shows the best performance in their experiments in NER in Brazilian legal documents.

Shi et al. [20] experimented with multiple combinations of models such as RoBERTa-biLSTM-CRF, Lattice-CRF, Lattice-LTSM and BERT-BiLSTM-CRF, while they have performance variations RoBERTa-BiLSTM-CRF outperformed all other architectures. In these architectures, the transformer models were used mainly to obtain rich embeddings. Duarte et al. [69] used BERTimbau a Brazilian-specific model in combination with a BiLSTM model. Overall, hybrid architectures mostly consist of BiLSTM, CRF which provided the best results in the pre-transformers era.

2.3 Legal NER shared task

Given the importance of NER in the legal domain, LegalEval2023[12] Sub-task B is designed to encourage novel approaches in legal NER, which help scholars to push the boundaries of legal NER systems collaboratively. The dataset they utilised was from Kalamkar et al. [77], and the same NER tags were used in the shared task, too. Seventeen teams participated in this Legal NER shared task, which shows great interest in this area.

As reported in the LegalEval2023 shared task overview paper [32], Ningthoujam et al. [124] were the best performing system for the NER task, where they proposed a Multi-Task Learning (MTL) framework. They augmented the data using back-translation as a solution to the class imbalance in the dataset. They achieved a 0.9120 F1 score for the NER task and completed the competition as the highest ranked system.

The second best system [125] managed to score 0.9099 F1 score, and they have used mainly two approaches (1) ensemble approach using multiple different transformer models in Spacy framework for predictions in combination with a weighted voting scheme. (2) adapted MultiCoNER (Multilingual Complex Named Entity Recognition) [126] with multiple pre-trained models to generate the contextualised features, which are further concatenated with dependency parsing features and part-of-speech features and were further passed through a CRF module for final predictions.

Cabrera-Diego and Gheewala [127] reported that the third best-performing system is based on FEDA (Frustratingly Easy Domain Adaptation) [128], and they trained their models on multiple legal corpora. The method extracted features from DeBERTa V3 [129] and passed them through two FEDA layers called general and specialised FEDA layers. The specialised FEDA layer also incorporated features from the general FEDA layer, and the final output was passed through a CRF module to generate the predictions. They achieved a 0.9007 F1 score, which ended up being third place in the competition (Table 2).

2.4 Legal NER summary and discussion

Section 2 presents different NER datasets and methods in the legal domain. We outlined 60 papers published in the past decade, which also included 30 datasets. The datasets have a good diversity as they are spread around multiple languages and jurisdictions, which is a positive sign in legal NER research.

One of the major limitations that we observe with the legal NER datasets is their availability. Only 12 out of 30 datasets are publicly available, making more than half of the legal NER datasets publicly unavailable. While we agree that this is a similar issue with other legal

[12] Available at https://sites.google.com/view/legaleval/home.

Table 2 Leaderboard results for LegalEval shared task in legal named entity recognition [32]

Rank	Team name	Reference	Model	Score
1	ResearchTeam_HCN	Ningthoujam et al. [124]	RoBERTa-base	0.9120
2	VTCC-NER	Tran and Doan [125]	RoBERTa + Transition based parser with dynamic weights	0.9099
3	DeepAI	NA		0.9099
4	Jus Mundi	Cabrera-Diego and Gheewala [127]	DeBERTa V3 + FEDA + CRF	0.9007
5	Autohome AI	NA		0.8833
6	uottawa.nlp23	Almuslim et al. [130]	LegalBERT	0.8794
7	TeamUnibo	Noviello et al. [131]	XLNet + BiLSTM + CRF	0.8743
8	DamoAI	NA		0.8627
9	AntContentTech	Huo et al. [132]	LegalBERT + CRF ensemble	0.8622
10	xixilu556	NA		0.8611
11	PoliToHFI	Benedetto et al. [133]	LUKE-base / BERT-base	0.8321
12	Ginn-Khamov	Ginn and Khamov [134]	RoBERTa + Sentence class token	0.7265
13	TeamShakespeare	Jin and Wang [135]	legal- LUKE	0.6670
14	UO-LouTAL	Bosch et al. [136]	SpaCy (preprocessing) + CRF	0.6489
15	Nonet	Nigam et al. [137]	Custom trained spaCy + BERT-CRF	0.5532
16	Legal_try	Zhao et al. [138]	ALBERT/RoBERTa + BiLSTM/IDCNN +CRF	0.5173
17	shihanmax	NA		0.0186

The reference column indicates the reference to the submitted shared task paper, while *NA* suggests that the team did not submit a paper

NLP tasks such as judgment prediction [39] and legal text summarisation [41], the scarce availability hinders further research and the impact of these datasets. Furthermore, five out of the 30 datasets that we presented do not describe the named entities [82–84, 88, 91], which further reduces the usability of the NER datasets. Figure 4 further illustrates that even the available datasets are majorly based on judgment text. The datasets for other document types such as legislative documents are limited as well as the languages.

We also described a number of methods experimented with legal named entity recognition, ranging from rule-based to recent transformer-based methods. Table 3 presents a comparative analysis of the legal NER systems. As evidenced by the recently concluded shared task, transformer-based methods provide state-of-the-art results. However, researchers have faced limitations in applying `LEGAL-BERT` [25] in NER tasks as it is an uncased model. Furthermore, since language-specific transformer models trained on legal corpora are limited, researchers have used transformer models trained on general corpora for legal NER tasks in other languages.

A common pitfall in existing legal NER methods is most of them have only been evaluated on one dataset. Researchers have often used their own datasets with specified annotation schemes to train and evaluate their machine learning models and have not experimented with different datasets. Therefore, we question the generalisability of the methods and the ability to perform in different datasets. This is also partially due to the fact that many datasets have not been publicly available. However, the ability to effectively generalise is consistently highlighted as a fundamental requirement for NLP models [150, 151]. Particularly in a real-world application such as legal NER, generalisation is crucial to ensure that the system exhibits robust, reliable, and fair behaviour when making predictions on data that differs from its training data. Therefore, the legal NLP community should be encouraged to perform comprehensive evaluations of the generalisability of legal NER methods and datasets.

3 *Task B:* legal relationship extraction

Relationship extraction (RE) refers to extracting semantic relationships between entities in a text. Depending on the sub-domain of law, the relationships can change in the context. Figure 6 shows an example of relationship extraction in drug-related criminal judgment documents. Relationship extraction is typically used in building knowledge graphs [152, 153], ontologies and understanding the key concepts of the text [154–156]. Recent surveys such as Detroja et al. [157] and Zhao et al. [158] provide a comprehensive view on RE data and methods in the general domain.

In the legal domain, RE becomes even more important for multiple reasons. Legal texts can contain multiple relationships, and they are critical in understanding legal situations. For example, if we consider the sentence, *"... On August 19, 2014, Mr. Su sold methamphetamine to Mr. Wang in Community A. ..."*, while Named Entity Recognition (NER) may flag entities such as Mr. Su (PER), Mr. Wang (PER), methamphetamine (DRUG), and August 19, 2014 (DATE), it falls short in understanding the connections between them. Legal professionals seek these connections to gain deeper insights into specific cases. Therefore, a system capable of extracting relationships such as *Mr. Su* **traffic_in** *methamphetamine* and *Mr. Su* **sell_drug_to** *Mr. Wang* proves invaluable, as it explains the entire situation for legal practitioners.

Relationship extraction holds significant importance across various legal documents, including legal contracts, court proceedings, and laws and regulations documents [159]. Iden-

Table 3 Legal NER methods

Method	Reference	Parameters	Tools/models	Code	Workflow
Lexicon and rule-based	Andrew [81]	–	–	No	Reg-ex/Pattern matching
	Solihin and Budi [87]	–	–	No	
	Thomas and Sangeetha [100]	–	–	No	
	Niu and Zeng [101]	–	–	No	
	Dozier et al. [22]	–	–	No	
	Andrew [81]	–	Wapiti toolkit [139], GATE [140], JAPE JAPE Thakker et al. [141]	No	CRF
	Sharafat et al. [89]	–	–	No	
	Glaser et al. [82]	–	CRFsuite [142], UIMA Verspoor and Baumgartner [143]	No	
	Glaser et al. [82]	–	DBpedia Spotlight Mendes et al. [144]	No	Software package
	Thomas and Sangeetha [100]	–	TIEx [145]	No	
	Glaser et al. [82]	–	DMP[a]	No	Template
	Kadu and Ashwini [99]	–	–	No	POS tags
	Schraagen et al. [91]		Frog NER[b]	No	
	Sharafat et al. [89]	–	–	No	Markov models
Machine learning-based	Chalkidis et al. [90]	–	LR, SVM	No	LR & SVM
Recurrent neural networks-based	Leitner et al. [106]	–	Combinations of LSTM, CRF and CNNs	No	LSTMs

Table 3 continued

Method	Reference	Parameters	Tools/models	Code	Workflow
	Nuranti and Yulianti [79]	–	Combinations of LSTM, CRF and CNNs	No	
	Luz de Araujo et al. [70]	Embedding dimension: 300, Char embedding dimension: 50, epochs: 55, dropout : 0.5, batch size: 10, optimiser: SGD, learning rate: 0.015, decay: 0.95, clipping: 5, 1st layer hidden units: 25, 2nd layer hidden units: 100	Combinations of LSTM, CRF and CNNs	No	
	Çetindağ et al. [73]	Embedding dimensions: 300, char level features dimension: 100	Combinations of LSTM, CRF and CNNs GloVe, Morph2Vec embeddings	Yes	
	Pais et al. [80]	Epochs: 100, early stopping epochs: 10	Combinations of LSTM and CRF, used NeuroNER [146], GeoNames and JRC-Names [147], CoRoLa [148], MARCELL[c]	No	
	Khasianov et al. [107]	–	Combinations of LSTM, CRF and CNNs	No	
	Wang et al. [113]	Batch size: 20, epochs: 20, LSTM dimension: 128, IDCNN dilation: [1, 1, 2], BERT fine-tune learning rate: 5e−4, learning rate of BiLSTM and IDCNN: 1e−4, dropout rate: 0.5	Combination of BERT, BiLSTM, ID-CNN and CRF	No	
	Chen et al. [88]	Neural network layers: 2, hidden layer LSTM neurons: 300, learning rate: 0.001, gradient clipping: 5, overfitting parameter: 0.5, batch-size: 64, Adam optimiser, epochs: 120	Combination of LSTM and CRF	No	

Table 3 continued

Method	Reference	Parameters	Tools/models	Code	Workflow
	Thomas and Sangeetha [100]	–	Combinations of LSTM, CRF and CNNs	No	
	Shi et al. [20]	Embedding size: 50, hidden size LSTM: 200, dropout: 0.5, optimiser: SGD, learning rate: 0.015, decay: 0.05, embedding size: 8, embedding learning rate: 2e−5	Combinations of RoBERTa, LSTM, CRF, Lattice-CRF and IDCNN-CRF	No	
	de Almeida et al. [85]	layers: 256, 128, 64, optimiser: SGD	LSTM	No	Seq-to-seq
Transformers	Ahmed et al. [21]	Epochs: 30, learning rate: 2e−4	`BERT-base-cased`, `BERT-base-uncased`, `LEGAL-BERT`	No	BERT-variations
	Duarte et al. [69]	Batch size: 32, epochs: 15, Bayesian optimiser	`BERTimbau`	No	
	Orasmaa et al. [74]	`EstBERT`—epochs: 9, batch size: 8, learning rate: 5e−05 `WikiBERT-et`—epochs: 10, batch size: 8, learning rate: 5e−05 `Est-RoBERTa`—epochs: 8, batch size: 16, learning rate: 5e−05	`EstBERT`, `Est-RoBERTa`, `WikiBERT-et`[d], CRF	Yes	
	Oliveira et al. [94]	Learning rate: 0.01, epochs: 10	`Lener-BR`, `BERTimbau`, `RoBERTa`, `DistilBERT-PT`	No	
	Tran and Doan [125]	Gradient accumulation: 3, 4, optimiser: adam, learning rate: 5e−5, 1e−4	`RoBERTa`, `BERT`, `LEGAL-BERT`, `InLegalBERT`, `XLM-RoBERTa`	Yes	
	Ningthoujam et al. [124]	Epochs: 4, learning rate: 5e−5, batch size: 4	`RoBERTa`, spacy transformers	No	

Table 3 continued

Method	Reference	Parameters	Tools/models	Code	Workflow
	Cabrera-Diego and Gheewala [127]	Epochs: 20, early stop patience: 2–5, learning rate: 2e−5, scheduler: linear with warm-up, warm-up ratio: 0.1, optimiser: Lookahead AdamW with bias correction, AdamWe: 1e−8, random seed: 12, dropout: 0.5, decay: 0.01, clipping gradient norm: 1.0, DeBERTa's sequence size: 256, 384 and 512, training batch: 32, context stride: 40	`DeBERTa V3 Large` [149]	No	
	Oleques Nunes. et al. [121]	Max token: 1000, cosine similarity threshold: 0.95	`Sabia-7B`	No	Generative models

Column *Method* refers to the high-level-method category. Column *Reference* provides the citation for the work, while column *Parameters* presents the model parameters used in their experiments. Column *Tools/models* contains the tools or the models used in the study, and column *Code* indicates if the source code of the experiments is released or not. If the code is publicly available, we have provided the link to the code repository as a hyperlink. Finally, column *Workflow* represents the specific methods employed in the study

[a] Available at https://github.com/google/diff-match-patch

[b] Available at https://languagemachines.github.io/frog/

[c] Available at https://marcell-project.eu/

[d] Available at https://huggingface.co/TurkuNLP/wikibert-base-et-cased

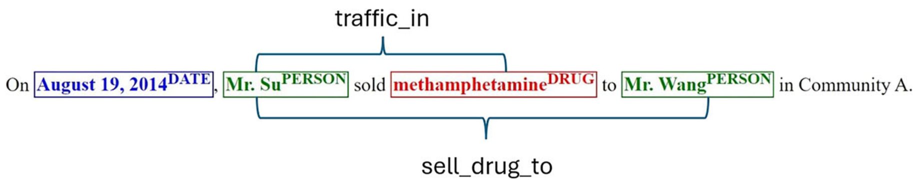


Fig. 6 Relationship extraction in drug-related criminal judgements

tifying these relationships manually is obviously inefficient, given the lengthy and complex nature of legal documents. Therefore, there is a substantial demand for automated systems capable of extracting relationships from legal documents.

As per the definition, we mainly focus on the semantic relations between named entities within the text. However, there are other research directions that consider relations between documents. A common example of this is considering citations of a court case as relations when court cases cite the related previous cases within their text. These types of relations help information retrieval systems largely and are reported by recent legal information retrieval surveys [46, 47]. Therefore, we have removed the papers related to document relations from our survey, as information retrieval is not our primary interest.

In the following sections, we describe the datasets and methods available for legal relationship extraction. Table 4 summarises the datasets.

3.1 Legal RE data

Similar to NER tasks, RE is also a supervised machine learning problem in which a machine learning model is trained on annotated data. Therefore, a properly annotated dataset is a main requirement in RE. Figure 7 illustrates the analysis of the RE datasets, which shows that they are limited to five main languages. This suggests the lack of resources for the RE task as well as the need to create the resources. One important observation is that the datasets are based on different legal domains, which makes it hard to compare and validate the new methods across different languages. It is noticeable that the datasets are fairly small in number of instances, and some of the datasets have not even mentioned their size. A complete detailed list of datasets can be found in "Appendix B".

3.2 Legal RE methods

Similar to the NER task, relationship extraction in legal documents is also explored using multiple methods, including rule-based [164] and neural methods [161]. Generally, RE approaches can be divided into two main categories: (1) pipeline-based RE approaches [165] and (2) joint RE approaches [166].

The pipeline-based approaches extract entities and relations through two separate stages, which first identify entities from the text and then detect the relation between them. Usually, the first stage uses NER methods to extract entities. The second stage classifies the relations of entity pairs. Pipeline-based RE approaches usually assume that the target entities are already identified, and the RE models only need to predict the relations between the pair of entities. However, these approaches are not common in the NLP community as they suffer from error propagation, where errors introduced during entity recognition impact the accuracy of relation classification [167].

Table 4 Legal RE datasets

Name/reference	Relationships	Language	Jurisdiction	Availability
Chen et al. [160]	traffic_in, sell_drug_to, possess, provide_shelter_for	Chinese	China	No
Andrew [81]	PERSONNE_FONCTION, PERSONNE_ROLE, SOCIETE_ROLE, SOCIETE_TYPE	French	France	No
Zhong et al. [161]	Pattern A1, Pattern A2, Pattern A3, Pattern A4, Pattern B	Chinese	China	No
Kurant [162]	cause-effect-relation	Polish	Poland	No
Schraagen and Bex [163]	residency	Dutch	Netherlands	No
Kwak et al. [93]	Coreference resolution, Beneficiary-Asset, Testator-Asset, Testator-Beneficiary, Testator-Will, Testator-Executor, Inclusion, Competence, Parent–Child, Witness-Testator, Witness-Will, Spouse-Spouse, Beneficiary-Executor, Beneficiary-Will, Testator-Codicil, Testator-Trustee, Testator-Guardian, Testator-Conservator	English	United States	Yes

Column *Name/reference* refers to the reference of the paper that introduced the dataset. Column *Relationships* refers to the annotated/considered relationships in the dataset, column *Language* shows the language used in the documents in the dataset, and column *Jurisdiction* refers to the jurisdiction the documents in the dataset belong to. The column *Availability* expresses whether the dataset is publicly available. If the dataset is publicly available, we have provided the link to the dataset as a hyperlink

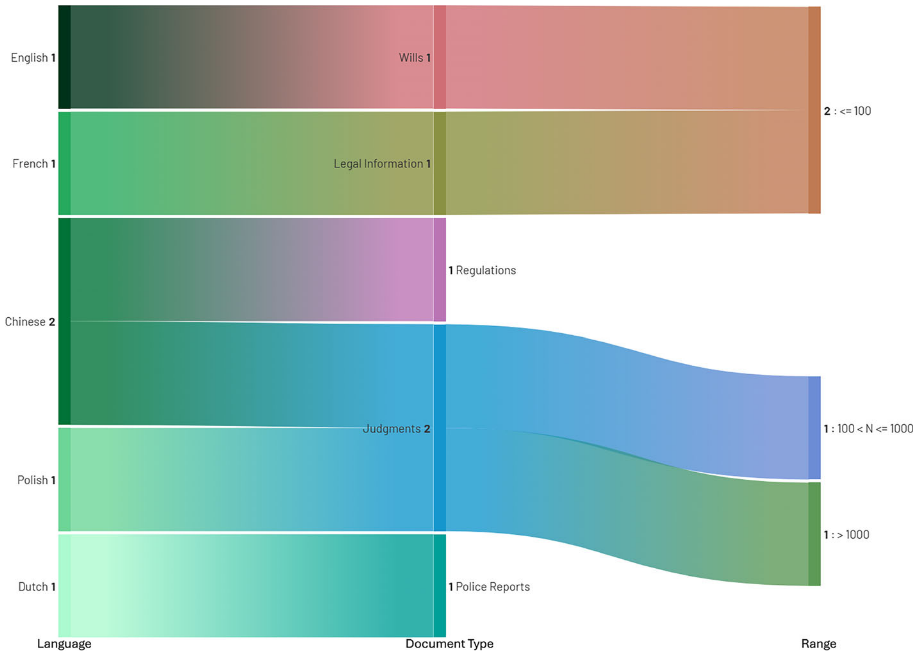


Fig. 7 Analysis of RE datasets by their language, domain and size. *N*—number of documents used

Joint RE approaches aim to perform entity recognition and relation classification simultaneously. Their output is usually triplets consisting of two entities and their corresponding relationships. Most joint RE approaches are neural-based.

An analysis of employed methods in RE is shown in Fig. 8. It is noticeable that transformers have not been employed much in the studies even though they deliver state-of-the-art results in many tasks in other domains. Only five languages have evaluated in terms of methods; however, the comparability of these studies is questionable based on the different types of documents they have used in their own studies.

3.2.1 Lexicon- and rule-based legal RE methods

Rule-based methods, in combination with ontologies, played a major role before the neural networks era and are still useful in use cases where training data is sparse.

Qady and Kandil [159] utilised CRISP (concept relation identification using shallow parsing) for concept relation extraction in construction documents. They use a pipeline-based approach where they first use rules to identify entities. At the next stage, they use Sundance [168] as an API to classify the relationship type. Their methodology demonstrated a notable enhancement over other systems at the time of publication.

Lame [169] extracted information from 57 different codes in French using Syntactical Analysis, Analysis of the Coordination Relations, Statistical Analysis and Pattern Matching. They have been able to extract the relations among legal terms successfully, and some of the above methods have needed manual overlooking after the generation of results, which could be considered as an additional overhead. This is also a pipeline-based approach.

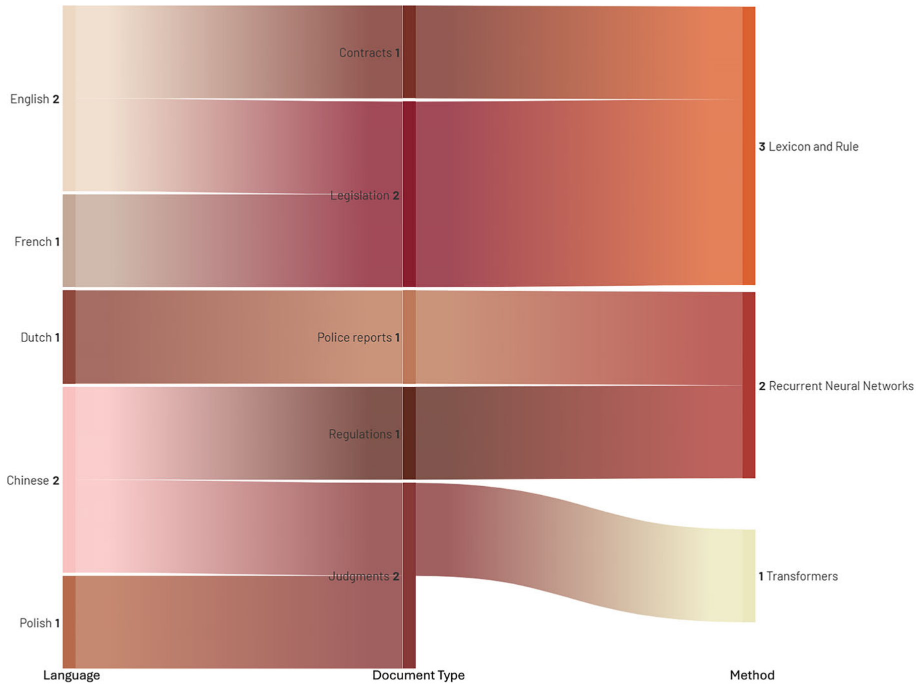


Fig. 8 Analysis of RE methods by their language, domain and method

Thimm and Schneider [170] created an NLP pipeline to extract relationships from Environmental Law. They employed LexNLP [171][13] as it is a powerful Python library implemented to process legal texts. As the name suggests, this is also a pipeline-based approach consisting of a sentence segmenter, a tokeniser, a part-of-speech tagger, an entity recogniser and finally, a relationship extractor.

All of the above rule-based methods are presented as tools/ software, and the authors have not conducted a quantitative evaluation on a dataset, which we also see as a major limitation.

3.2.2 Deep learning-based legal RE methods

Similar to NER methods, LSTMs and BiLSTMs have been used commonly in relationship extraction tasks, too. They have provided good results with many RE datasets.

Zhong et al. [161] used `BiLSTM-CRF` and `LSTM-MLP` for effective extraction of construction procedural constraints from construction regulations in Chinese legal documents. Specifically, `BiLSTM-CRF` has been used for Named Entity Recognition, while the `LSTM-MLP` was used for relationship extraction. They used national standards of the Code for Acceptance of Construction Quality in China, which the text is in Chinese. They report good results ranging from the lowest f1 score of 0.571 to and highest f1 score of 0.845 for the identification of constraint patterns.

Kurant [162] have used LSTMs to extract "cause-and-effect" relationships in polish court judgements. With their 2 network architecture, the first network identifies if a sentence has a causality or not, and the second is to extract the cause and effect parts from the sentence.

[13] Available at https://lexpredict-lexnlp.readthedocs.io/en/latest/.

From the second network, the output is a classification of each word extracted according to the following classes: cause (class 0), effect (class 1), causal phrase (class 2), and other (class 3). They show they could achieve 0.58, 0.51, 0.80 and 0.09 respective F1 scores for the above mentioned classes.

Schraagen and Bex [163] described a study on relationship extraction in Dutch police reports. They compared LTSM and SVM models models having two separate models from each architecture and two separate sets of data of the same dataset, and they have cross-validated the results. In all cases, LSTMs perform better, showing the power of neural architectures over traditional machine learning models such as SVMs.

Finally, Chen et al. [160] introduced NER and relationship extraction dataset in Chinese drug-related criminal judgment documents, while they utilise `RoBERTa` [172] encoder and decoder in combination with the feature enhancement method. This is the only joint entity and relation extraction method we identified that was published for the legal domain. As reported, their approach achieves an F1 score of 0.836 in triplet extraction, demonstrating the high effectiveness of pre-trained transformer models.

3.3 Legal RE summary and discussion

Relationship extraction is one of the key stages of an NLP Information Extraction pipeline. Despite the popularity of Legal NER within the community, relationship extraction remains relatively unexplored. As can be seen in Table 4, only 6 datasets have been explored for legal RE. One possible reason for the unpopularity of RE in the legal domain is the high cost of annotations. To annotate legal relationships, annotators must first annotate the named entities and then define the relationships among them. This process is time-consuming and requires legal expertise for successful completion. While the general domain utilises techniques such as distant supervision [173] in order to get rid of manual annotations, only one out of the six legal relationship extraction datasets uses distant supervision. We suggest that the LegalNLP community should utilise more techniques similar to distant supervision [173] to improve RE in legal texts. Additionally, the issue highlighted in legal NER regarding the public accessibility of the datasets also applies to legal relationship extraction, as only one dataset is publicly available.

We also described a number of methods experimented with for legal relationship extraction ranging from rule-based to neural models. Table 5 provides a summarised view of legal RE methods. We identify several limitations with them: (1) All the methods have been experimented with mostly one dataset, and some of them are not even evaluated on one dataset. This restricts the generalisability of these methods. (2) State-of-the-art methods in the general domain, such as transformers with joint entity and relation extraction, remain underexplored in the legal domain. As information extraction pipelines are not complete without a proper relationship extraction stage, the legal NLP community should focus more on legal RE.

4 *Task C:* legal event detection

Event Detection refers to detecting the appearance of an event in the text (e.g. sentence, document) and its related information mentioned in the provided text. An event is an activity that takes place at a specific time and location, or it can be described as a transition from one state to another [174, 175]. The following terms have been commonly used in event

Table 5 Legal RE methods

Method	Reference	Parameters	Tools/models	Code	Workflow
Lexicon and rule-based	Qady and Kandil [159]	–	Sundance	No	Rules/pattern matching
	Lame [169]	–	–	No	
	Thimm and Schneider [170]	–	LexNLP [171]	No	Software package
Deep learning-based	Zhong et al. [161]	Input layer units: 50, hidden layer units: 40, output layer units: 5	LSTM-MLP	No	LSTMs
	Kurant [162]	Embedding dimentions: 100, first network epochs: 8, second network epochs: 10	BiLSTMs	No	
	Schraagen and Bex [163]	Sequence length: 21 tokens, input data type: integer vocabulary size: 4938, lemma: 4151, embedding layer vector length: 512 mask zero: true, LSTM layer units: 32 merge: concatenate, auxiliary output type: dense, activation: sigmoid, hidden layers type: dense, output units: 64, hidden layer activation: rectified linear unit, number of hidden layers: 3, optimiser: root-mean-square propagation, loss function: binary cross entropy	LSTM, SVM	No	
	Chen et al. [160]	–	RoBERTa, Combinations of BERT and LSTMs	No	BERT-variations

Column *Method* refers to the high-level method category. Column *Reference* provides the citation for the work, while column *Parameters* present the model parameters used in their experiments. Column *Tools/models* contains the tools or the models used in the study and column *Code* indicates if the source code of the experiments is released or not. If the code is publicly available, we have provided the link to the code repository as a hyperlink. Finally, column *Workflow* represents the specific methods employed in the study extraction. Based on these terms, Ahn [176] proposes to divide the event extraction into sub-tasks: event sentence identification, trigger detection, argument detection, and argument role classification.

- *Event extent* is a sentence within which an event is expressed. Event extent detection is typically considered as a text classification task [177, 178].
- *Event trigger* is a word or phrase that most distinctly indicates the occurrence of the event. Event trigger identification is typically considered as a token classification task [179].
- *Event arguments* are entities that are part of the event, which include participants and attributes [180, 181]. Event argument identification is also considered as a token classification task.
- *Argument role* is the relationship between an event and its arguments [182–184].

These terms are used to describe an event. Therefore, an event detection pipeline involves identifying the occurrence of an event and extracting various properties, such as triggers and arguments of a particular event. While some studies address the entire event detection process, others concentrate on specific steps within the pipeline. Recent surveys like Li et al. [12] show the progress of event detection in the general domain.

Events are important in many sub-domains of law, such as Criminal Law, Civil Law, Corporate Law and Immigration Law. Events carry valuable information about actions and behaviours. Typically, this information is significant in a court trial. The judges carefully consider the events that happened in the situation before deciding the verdict. Similarly, level event detection (Legal ED) can help with Legal Judgment Prediction [185] and similar case identification tasks [186, 187].

While the named entities or their relationships can be event arguments and argument roles in a particular event, a NER method or a RE method, let alone, cannot be used to describe an event. For example, in *Alice hit Bob with his car on the road and escaped*, while a NER method will identify Alice and Bob as PERSON; it cannot identify Alice and Bob as the event arguments and argument roles.

We identify the need for event detection tasks in legal applications, especially in judgment predictions and similar case retrievals [186]. The following sections describe the datasets and methods released for legal event detection.

4.1 Legal ED data

Similar to the previous tasks we described, researchers have introduced several datasets for legal event detection that are suitable for training supervised machine learning models. Table 6 summarises the datasets.

Figure 9 shows the analysis of the datasets in ED. A large portion of datasets are Chinese, and they are all based on judgment texts. Except for one dataset in English based on wills, all other datasets available are on judgments, which makes the datasets less varied and emphasises the lack of resources for the community. A complete, detailed list of datasets can be found in "Appendix C".

4.2 Legal ED methods

Similar to Tasks A and B, different methods, including rule-based and neural methods, have been implemented to perform legal event detection. Figure 10 shows that the limited avail-

Table 6 Legal ED datasets

Name/reference	Number of event types	Number of event arguments	Number of argument roles	Language	Jurisdiction	Availability
Yao et al. [188]	108	–	–	Chinese	China	Yes
Li et al. [189]	5	6	5	Chinese	China	No
Shen et al. [190]	11	26	17	Chinese	China	No
Li et al. [191]	13	–	–	Chinese	China	No
Filtz et al. [192]	2	–	–	English	Europe	Yes
Paulino-Passos et al. [193]	7	–	–	English	United States	Yes
Araujo et al. [194]	4	–	–	Portuguese	Brazil	No
Kwak et al. [93]	20	–	–	English	United States	Yes
Xian et al. [195]	7	9.9 (on average)	378	Chinese	China	No

Column *Name/reference* refers to the reference of the paper that introduced the dataset. Column *No. of event types* refers to the number of Event Types used in the dataset, column *No. of event arguments* refers to the number of Event Arguments used in the dataset, column *No. of argument roles* refers to the number of Argument Roles used in the dataset, column *Language* shows the language used in the documents in the dataset, column *Jurisdiction* refers to the jurisdiction the documents in the dataset belongs to. The column *Availability* expresses whether the dataset is publicly available. If the dataset is publicly available we have provided the link to the dataset as a hyperlink

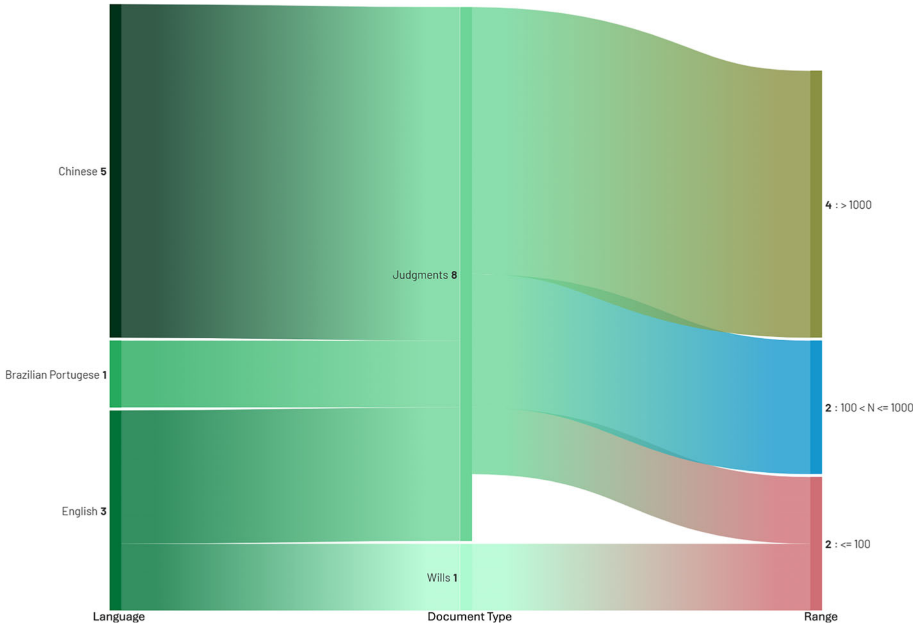


Fig. 9 Analysis of ED datasets by their language, domain and size. *N*—number of documents used

ability of methods for different languages and only three languages have been experimented in ED. Moreover, ED has been limited only to judgment texts, and interestingly, the transformers are yet to be explored in this space. This suggests that ED is heavily understudied and need more research with different languages and different methods in this area.

4.2.1 Lexicon- and rule-based legal ED methods

Araujo et al. [194] studied legal event extraction in Portuguese in Brazilian legal documents. They employed a two-phased methodology in which, firstly, they conducted a corpus study to build a domain ontology and linguistic rules. This exercise resulted in twelve linguistic rules targeted at extracting the 4 events we mentioned in the data section. Then, the following phase utilised the crafted linguistic rules and the ontology as inputs to the SAURON inference system to extract the events. Their experimental results show higher precision and recall values for all four event types, 0.84 being the lowest recall, which resulted for Conviction and 1.00 being the highest recall value, which is for the Questioning event type. Araujo et al. [196] followed a similar approach on different Brazilian legal text data to extract the same event types as we described in the data section [194].

4.2.2 Deep learning-based legal ED methods

LSTMs and transformer models have been used commonly in legal event detection tasks. They have provided state-of-the-art results in many datasets.

Yao et al. [188] evaluated different neural network models on Chinese criminal law event detection. First, they used BiLSTM and BERT as their baseline models. Then, they extended their methodology to dynamic max-pooling (DM)-based models such as `DMCNN` [197] and

DMBERT [198] to extract the sequence features and employ dynamic pooling layers to obtain trigger-specific representation. Finally, they experimented with sequence labelling models such as BiLSTM+CRF and BERT+CRF. They have evaluated their models on the LEVEN dataset we mentioned before. DMBERT shows the highest results with an 0.8034 macro F1 score. However, even though the best model is DMBERT, they have used BERT+CRF model for their subsequent tasks; legal judgement prediction and similar case retrieval due to the complexity of DMBERT. BERT+CRF also shows competitive performance in event detection.

Li et al. [189] experimented with event extraction in larceny law, again on Chinese legal documents. They used BERT to obtain character-level vector representations and then used a BiLSTM-CRF layer for the token-level classification task. The output of the BiLSTM-CRF layer is a sequence of word-level BIO tokens relevant to the event types we described in the data section. They jointly extract the event arguments in this step. They compared their event extraction with a baseline model called a two-labelling model. In the first step, a combination of word2vec, BiLSTM and POS tagging, was used to extract the transition label. Then, the embedding of transition label, trigger word and positional embedding as input to a CRF model to extract the event type. Their experiments report 0.8466 and 0.8553 F1 scores for the baseline and their method, respectively.

Li et al. [191] also used a Bilstm in combination with CRF in their experiments on divorce cases in China, but they used a trigger-words dictionary to filter the sentences which are needed to be used in the event detection experiment.

Filtz et al. [192] studied event extraction using three main approaches. Namely, they are deep learning-based, such as Bert [24], Distilbert [199] language models, CRF-based and rule-based approaches. Their results show that the pre-trained language models are clearly ahead. The Distilbert finetuned model reported 0.9238 and 0.7975 F1 scores for Procedure and Circumstance event types, respectively, which outperformed CRF and rule-based methods by at least 0.10 F1 score in both events.

[200] studied event extraction in European legal decisions. They used a structure extracted to first identify the document structure and send the sentences of the important parts of the documents to extract events. They created frames for representing the main verbs found in the sentences and grouped the sentences containing the particular verb. And they have selected the events based on temporal expressions, if there is any, they have selected the event resulting a list of events. Their results show that this approach could achieve a 0.8477 F1 score in event identification, while their baseline could achieve only a 0.8163 F1 score.

Joshi et al. [201] used event extraction in Indian court cases for the task of prior case retrieval. They rather follow an automatic approach for event extraction using dependency parsing with Spacy. They used the verbs as the root dependency and checked the left and right of them to parse the events.

Different to well-known neural models such as BiLSTM [202] and BERT [24], Shen et al. [190] proposed a novel architecture called Pedal Attention-based Joint Hierarchical Event Extraction (PAJHEE). They have six modules in their architecture responsible for extracting events and event types, building the hierarchical event type features, computing pedal attention, extracting trigger, argument role prediction and finally, the joint inference module. Their experiments were on Chinese legal documents. They evaluate and show that their architecture is capable of achieving the superior performance of 0.935 F1 score, surpassing the benchmark models; DMCNN [197], DBRNN [203] and PLMEE [204] on the event identification task.

Xian et al. [195] used deep learning-based methods to create baselines for the document-level event extraction dataset they introduced. They used 3 benchmark models, BERT_QA [205], PTPCG [206], LIC_DEE [207]. As these models are not specific to legal knowledge, the authors have further extended their experiments to enhance the embedding layers of the

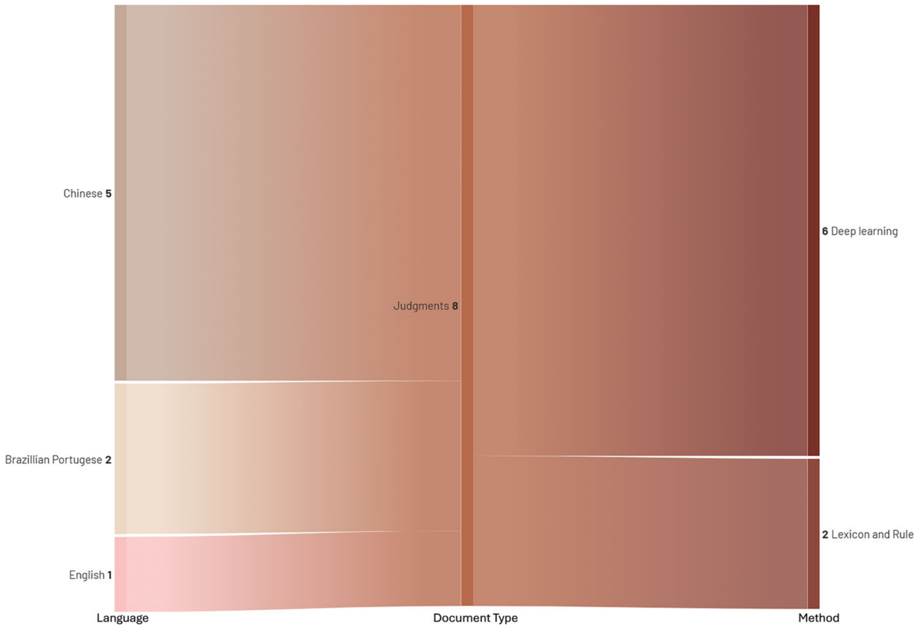


Fig. 10 Analysis of ED methods by their language, domain and method

above models using Criminal Document Bert [208]. Their results show BERT_QA (legal) performed best, reporting a 0.9587 F1 score, followed by the PTPCG (legal) model with a 0.9482 F1 score.

4.3 Legal ED summary and discussion

In this section, we presented different legal ED datasets and methods in the legal domain. We outlined 19 papers published in the past decade, including 9 datasets.

The existing literature suggests that legal event detection can provide basic building blocks for similar case retrieval [201] and judgment prediction tasks [211]. In similar case retrieval, people extract the events that happened in the source case and through vector-based modelling or language model-based modelling, they see the similarity between the target and source cases in terms of the events that happened. Similarly, events have been used as a feature in legal judgment prediction systems along with other features like fact-based features and contextual features.

As summarised in Table 6, there were only 9 datasets for legal event detection. The datasets do not have a good diversity as they are only available in English and Chinese, which we believe is a major limitation in Legal ED datasets. Furthermore, only two datasets cover all the tasks in ED, i.e. event argument extraction and event argument roles extraction. Finally, similar to *Task A and B*, less publicly available data is a limitation in legal ED, too.

We also described a number of methods experimented with legal event detection, ranging from rule-based to recent neural-based methods. Table 7 illustrates the different methods used in the legal ED literature. Transformer-based methods provide state-of-the-art results in many datasets. Interestingly, LLMs such as GPT and Llama have not yet been used in legal ED. Furthermore, similar to *Tasks A and B*, most of the methods have only been evaluated using

Table 7 Legal ED methods

Method	Reference	Training parameters	Pre-trained models	Code	Workflow
Lexicon / Rule-based	Araujo et al. [194]	–	POWLA Chiarcos [209], SAURON	No	Ontology/Rules/SAURON package
	de Araujo et al. [196]	–	POWLA Chiarcos [209], SAURON	No	
Deep learning-based	Yao et al. [188]	DMCNN: {Batch size: 170, dropout rate: 0.5, learning rate: 1e−3, kernel size: 3, hidden size: 200, dimension of PF: 5, dimension of embeddings: 300}, BiLSTM: {Batch size: 200, dropout rate: 0.5, learning rate: 1e−3, kernel size: 3, hidden size: 256, dimension of embeddings: 300}, BERT: {Batch size: 64, dropout rate: 0.5, adame: 1e−8, learning rate: 5e−5, validation steps: 500}	`BiLSTM`, `BERT`, `DMCNN`, `DMBERT`, Combinations of BERT, BiLSTM with CRF	Yes	LSTMs
	Li et al. [189]	–	`BiLSTM-CRF`, `BERT`	No	
	Li et al. [191]	–	Combination of `BiLSTM` and CRF	No	
	Filtz et al. [192]	–	`BERT`, Flair [210], `DistilBERT`, CRF	Yes	BERT-variations
	Shen et al. [190]	Argument category and dependency syntax dimension: 100, hidden layer size: 256, gradient descent over shuffled mini-batches, learning rate: 0.00001	Pedal attention	No	

Table 7 continued

Method	Reference	Training parameters	Pre-trained models	Code	Workflow
	Xian et al. [195]	BERT_QA: {Batch_size: 8, learning_rate: 0.001, optimizer: Adam gradient_accumulation_steps: 5, warmup_steps: 0, hidden_size: 768} PTPCG: {Batch_size: 96, learning_rate: 0.001, optimizer: Adam gradient_accumulation_steps:8, dropout: 0.1, hidden_size: 768} LIC_DEE: {Batch_size: 4, learning_rate: 0.0005, optimizer: Adam gradient_accumulation_steps: 8, warmup_proportion: 0.1, hidden_size: 768}	BERT_QA [205], PTPCG [206], LIC_DEE [207]	No	

Column *Method* refers to the high-level method category. Column *Reference* provides the citation for the work, while column *Parameters* presents the model parameters used in their experiments. Column *Tools/models* contains the tools or the models used in the study, and column *Code* indicates if the source code of the experiments is released or not. If the code is publicly available, we have provided the link to the code repository as a hyperlink. Finally, column *Workflow* represents the specific methods employed in the study one dataset, which raises questions about the generalisability of these methods. Moreover, event detection can be extended to many sub-domains of the law, such as Civil, Environmental, and Intellectual law. We reckon the existing legal ED research has not covered these other law areas.

5 Open challenges and future directions in legal IE

This section presents the current issues of legal information extraction and the potential future directions from an NLP perspective.

Data scarcity Performances of the state-of-the-art deep learning IE models heavily depend on the training dataset. However, as we mentioned before, all the IE tasks suffer from data scarcity. The majority of the datasets we presented in this paper are not publicly available. Out of 45 datasets we discussed, only 18 are available online, which is only 40%. Moreover, Figs. 4, 7 and 9 clearly illustrate that even the available datasets are heavily based on one legal document type which is judgments. Datasets for other document types like contracts and legislation are heavily under-developed. A great portion of available datasets have less than 100 data instances, which further limits the applicability of the state-of-the-art deep-learning and transformer models.

However, data scarcity demotivates researchers who work on the topic and restricts research developments. Therefore, *LegalNLP* community should actively seek solutions for data scarcity as a future research direction. The community should collaborate with legal database firms and research possible data anonymisation techniques so that the datasets can be released publicly without breaching privacy concerns [212, 213].

Model generalisability The legal domain is generally lacking benchmark datasets for information extraction. As we mentioned before, while there exists a benchmark for legal language understanding, it does not contain any IE tasks [56]. Therefore, most researchers work on their own datasets for model evaluation and do not consider other datasets. We question the generalisability of the methods in terms of the ability to perform in different datasets [214]. While there are several efforts to improve the generalisability of NLP methods, such as GenBench workshop [215],[14] none of them are available for the legal IE domain. Figure 5 indicates that almost all of the NER research have used their own datasets for their research. They have not utilised the existing datasets even if they are in the same category such as judgments. This is because different studies define their desired NER tags and those are slightly different from the other existing datasets. However, we identify making benchmark datasets along with tag sets can largely help the research specially in transfer learning when there is an overlap of tags. The same pattern is visible in the other IE tasks as well. Generalisability is one of the primary desiderata for models of natural language processing [215], especially in an applied topic like legalIE. Therefore, we see developing IE benchmarks and generalisable IE methods as crucial in order to transfer knowledge from research work to practical applications.

Multilinguality and Multijurisdictionality The majority of the research we presented in this survey is based on high-resource languages such as English or Chinese and high-resource Jurisdictions such as the USA. As there are approximately 190 jurisdictions, legalIE research should also focus on less-resourced languages and jurisdictions in terms of developing datasets as well as methods. The researchers should utilise techniques such as cross-lingual [216, 217] and cross-jurisdiction [218, 219] transfer learning to facilitate low-resource

[14] Available at https://genbench.org/workshop/.

research. Even though there are multiple datasets in one document type for an example Judgments in Fig. 9, the employed methods are largely different and do not utilise any sort of common frameworks to merge the multilingual research. Unlike other research domains, there are very less or no efforts on creating multilingual, multijurisdictional legal models which could largely support the multilingual legal research.

Complete IE pipeline research In this survey, we discussed all the steps in an IE pipeline: named entity recognition, relationship extraction and event detection [5]. Most of the legal IE research we observed focused only on one or two pipeline stages. To the best of our knowledge, only [93] consider all three IE pipeline stages. Furthermore, comparatively, legalNER has attracted more research than legalRE and legalED. Complete IE pipeline research can hugely benefit the practical applications. Therefore, researchers should focus more on the complete IE pipeline in the future.

Employing large language models Large language models have gained increasing attention in the NLP community as they have delivered state-of-the-art results in many NLP problems, including IE [220–222]. However, as we observed throughout our survey, it was not common for the researchers to use large language models in legalIE except for a handful of research [93, 94, 121]. In future work, we see large language models getting employed in legalIE in two ways;

1. *State-of-the-art models* As shown by IE research in general domain [223, 224], large language models have provided state-of-the-art results in IE tasks. Importantly, they do not need a large number of annotated instances as they have been used through zero-shot [225–227] or few-shot prompting [228, 229]. Therefore, legalIE can benefit from large language models, receiving state-of-the-art results from a limited number of training instances.
2. *Data annotators* Annotating legalIE data using humans can be very expensive, considering the fact that most of the annotators are legal practitioners [230, 231], which is also a reason for data scarcity in legalIE. Zhang et al. [232], Bogdanov et al. [233], Meoni et al. [234] have used large language models as annotators to create training data for different tasks in IE. Therefore, large language models should be explored as data annotators in legalIE in future work.

Ambiguities and context dependency Legal domain consists of many areas that are being explored under LegalNLP. Case-law, legal contracts, legal wills, legislation documents, terms and conditions are few popular examples. Inherently, these different areas carry specific meanings for specific terms. For an example, parties in a legal contract and parties in a judgment are semantically different and specific to the area of the law. Understanding these nuances is not a trivial task, specially in the fine-grained information extraction. These complexities are reflected in the current research landscape as each of the available legal IE datasets focuses only one specific legal area. This reduces the noise and isolates the problem. However, the future trends like large language models do not treat domains separately which demands thorough future research in this area.

6 Conclusions

In this paper, we conducted a systematic review to identify methods and datasets for legal information extraction. Overall, we discussed 83 papers covering all three stages in an IE pipeline: named entity recognition, relationship extraction and event detection. We noted that much attention has been given to the development of NER approaches, and the other two stages remain underexplored. In each stage, we listed the datasets we came across, discussing their properties, such as annotated concepts, size, language jurisdiction, and availability. We also discussed the available methods and how they have been evaluated. At the end of each section, we presented a discussion on the strengths and weaknesses of available datasets and methods. Finally, at the end of the survey, we identified open challenges in legalIE and made recommendations for future research, mainly in the directions of data scarcity, generalizability of the methods, multilinguality, multijurisdictionality and opportunities with LLMs.

With this paper, we presented a complete and up-to-date survey available for legalIE. Alongside the paper, we also released a web portal[15] containing all the papers we discussed in this survey. We believe that the survey and the web portal will be useful for both computer scientists and legal practitioners who are interested in information extraction in the legal domain.

Appendix A: Overview of legal NER datasets

In the following list, we describe the available legal NER datasets released in the last decade.

Court judgments and case law datasets

Ahmed et al. [21] dataset consists of publicly available judgments of the courts in Pakistan. They have first used the dataset in Sharafat et al. [235], which is on civil judgments in Lahore High Court[16] (LHC). This dataset has 100 cases, and they have split the training and test sets 90% and 10%, respectively. Secondly, they used a new dataset of Civil Appeal judgments in Supreme Court Pakistan[17] (SCP), which contained 214 judgements and they used 80% 20% splits between training and testing. Both of the datasets are in English. They have not made their annotated datasets publicly available. The combined dataset contains fourteen named entities in judgments related to "civil appeal" cases, which are described below.

1. **Per**—Name of a person; including judges, lawyers and others involved
2. **Loc**—Any location, name of a place or address
3. **Org**—Name of an organization or business
4. **CaseNo**—Assigned case number of the case being decided in the judgment
5. **Resp**—Name of respondent in the case be it an organization or a person
6. **Date**—Any complete date mentioned
7. **Refcourt**—Court that passed a referred judgment
8. **Refcase**—Any past judgment being referred to
9. **Ref**—Law references mentioned
10. **Appealcourt**—Court that passed the judgment that is being appealed against
11. **Appealcaseno**—Case number of the case being appealed against in the current case
12. **Money**—Any monetary amount mentioned, includes fines, debts etc.
13. **FIRno**—First Investigation Report (FIR) number that might be filed with the police
14. **Approved**—Whether the judgment is approved for reporting

[15] The anonymised link to the web portal is available at https://anonymous.4open.science/r/legal-information-extraction-8F2F/. Once the paper is accepted, we will add the original link to the web portal.

[16] Available at https://www.lhc.gov.pk/.

[17] Available at https://www.supremecourt.gov.pk/.

Leitner et al. [71] present a dataset that contains approximately 67,000 sentences in German legal documents. The dataset contains 54,000 annotated entities on 19 semantic concepts. Court decisions from 2017 and 2018 were selected for the dataset, published online by the Federal Ministry of Justice and Consumer Protection.[18] The authors have not specified the training and the testing splits. The dataset is publicly available at their GitHub repository.[19]

1. **PER**—Person
2. **LOC**—Location
3. **ORG**—Organisation
4. **NRM**—Legal norm
5. **REG**—Case-by-contract regulation
6. **RS**—Court decision
7. **LIT**—Legal literature

LeNER-Br [70] report a dataset dedicated to NER in Brazilian Portuguese. They created this dataset by collecting 66 legal documents from multiple Brazilian courts, a mix of superior and state-level courts. They have further added four legislation documents, such that 70 documents are in the dataset. Training, dev and test datasets consist of 50, 10, 10 documents, respectively. The dataset is not publicly available. The following list shows the Named Entities their study aimed at.

1. **PESSOA**—Person
2. **TEMPO**—Time
3. **LOCAL**—Location
4. **JURISPRUDENCIA**—Decisions of Legal cases
5. **ORGANIZACAO**—Organisation
6. **LEGISLACAO**—Laws

Çetindağ et al. [73] compiled their dataset using a publicly available database called Legalbank[20] and used the cases from the Turkish Court of Cassation in their study. Annotated 350 court cases, which consisted of 5311 named entities. The full dataset contains 2198 sentences, and they have used 1758 and 439 sentences as their training and test sets, respectively. The dataset is available at their GitHub repository.[21]

1. **PER**—Person
2. **LOC**—Location
3. **ORG**—Organisation
4. **DAT**—Date
5. **LEG**—Legislation
6. **COU**—Court
7. **REF**—Reference
8. **OFF**—Official Gazette
9. **O**—Other

Brugman [75] dataset consists of two main datasets. The first is 1000 randomly selected cases from 2017, while the second is 1000 randomly selected cases by the Supreme Court of the Netherlands from 2017. In both datasets, the authors used 400, 100, and 500 cases for training, development, and test sets. They created the dataset using the Council for the judiciary publishes judgements.[22] The authors have not made their data publicly available. Following are the list of unique named entities in their dataset.

[18] Available at https://www.rechtsprechung-im-internet.de.

[19] Available at https://github.com/elenanereiss/Legal-Entity-Recognition.

[20] Available at https://legalbank.net/arama.

[21] Available at https://github.com/BerkayYazicioglu/LegalNER.

[22] Available at https://www.rechtspraak.nl/.

1. **CASE**—Case references
2. **LOC**—Locations
3. **LEG**—Courts
4. **ID**—Non-standardised number
5. **DATE**—Dates
6. **SECTION**— Section headers
7. **NJ**—Dutch Case Law identifier
8. **ECLI**—European Case Law Identifier
9. **PRISON_SENTENCE**— the type of incarceration, imprisonment or juvenile detention
10. **FINANCIAL_SENTENCE**—the legal ground for a financial punishment, such as property damages or a fine
11. **DURATION**—a time span
12. **CURRENCY**—is used to annotate all amounts in euros
13. **PROBATION**—the probation period
14. **CONVERSION**— the conversion rate

Kamat et al. [76] introduced an annotated Indian Court Decision Document Corpus consisting of ten coarse-grained classes and 30 fine-grained classes. The sentences have been scraped from the Indian Kanoon website.[23] They have made the dataset publicly available via their GitHub.[24] The authors have not mentioned the training and test splits in their paper. The ten coarse-grained classes are described below.

1. **CRT**—Court
2. **PT**—Legal Party
3. **CD**—Court decision
4. **DOC**—Legal document
5. **JURD**—Jurisdiction
6. **LOC**—Location
7. **CTYP**—Case Type (Civil, Criminal)
8. **AUTH**—Judge who delivers judgment
9. **CRTOF**—Court officials
10. **DOJUD**—Date of judgment

Kalamkar et al. [77] provide another NER dataset in Indian court documents, which are also scraped from Indian Kanoon. The corpus consists of 14,444 Indian court judgment sentences and 2126 judgment preambles annotated with 14 named entities. In judgment sentences, authors split 9435, 949, and 4060, respectively, among training, development and test sets, and in judgment preambles, these splits are of 1560, 125, and 441 samples. They have made their data publicly available on GitHub repository.[25] Following are the named entities they experimented with.

1. **COURT**—Name of the Court
2. **PETITIONER**—Name of the petitioners
3. **RESPONDENT**—Name of the respondents
4. **JUDGE**—Names of the judges
5. **LAWYER**—Names of the lawyers
6. **DATE**—Any date
7. **ORG**—Name of organisations
8. **GPE**—Geopolitical locations
9. **STATUTE**—Name of the act or law
10. **PROVISION**—Sections, sub-sections, articles under the statute
11. **PRECEDENT**—All the past court cases
12. **CASE_NUMBER**—All other case numbers except for precedent
13. **WITNESS**—Name of witnesses
14. **OTHER_PERSON**—Name of all the persons

Nuranti and Yulianti [79] dataset consists court decisions of Indonesian supreme courts. The dataset consists of 1000 court decisions from five different courts in Indonesia. The dataset

[23] Available at https://indiankanoon.org/.

[24] Available at https://github.com/semintelligence/KING/tree/main/pre-processing.

[25] Available at https://github.com/Legal-NLP-EkStep/legal_NER.

features ten named entities, as described below. The authors have used 90% of the data as their training instances, and the rest, 10%, they have used for testing. The dataset is not publicly available online.

1. **advokat (ADVOCATE)**—Person who provides legal services
2. **amar (PUNISHMENT)**—Punishment of a court decision
3. **hakim (JUDGE)**—Justice official
4. **jaksa (PROSECUTOR)**—Government officials
5. **organisasi (ORG)**—Organisation
6. **panitera (REGISTRAR)**—Person who makes treatise
7. **peraturan (LAW)**—Legal rules
8. **putusan (DECISION_NO)**—Decision call number
9. **tanggal (DATE)**—Date
10. **terlibat (PER)**—A defendent, applicant, witness, respondent and plaintiffs

Naik et al. [78] report an annotated dataset of 30 Indian court documents collected online[26] resulting approximately 5000 sentences with eight named entities. The authors have not mentioned the training and test split percentages of the data, nor have they made the data publicly available.

1. **PERSON**—Name of the person
2. **LOC**—Locations which include names of states, cities, villages
3. **DATE**—Any date mentioned
4. **ORG**—Name of organisation
5. **Court**—Name of the court
6. **LEGAL**—Sections, sub-sections, articles orders etc
7. **ACT**—Act name in constitution
8. **CASE_NO**—Case no. of court judgments

Glaser et al. [82] utilised two types of data in their experiments: 1. Randomly selected corpus of 20 judgments from the law of tenancy of the 8th Zivilsentat of the German Federal Court of Justice.[27] 2. Five legal contracts in German. Both datasets used the same named entities. We have listed the named entities below, but they have not provided a description for each named entity tag. The datasets are not publicly available.

1. **PER**
2. **ORG**
3. **LOC**
4. **DA**
5. **MV**
6. **REF**
7. **OTH**

Krasadakis et al. [83] manually annotated over 4000 paragraphs from over 150 Greek legal documents for the task of named entity recognition. They have not provided details about the training and test splits. They used eight named entity types, which are listed below, but they have not provided a description for each named entity. Neither did they make their dataset freely available.

1. **LAW**
2. **CASE LAW**
3. **EURO LAW**
4. **EURO CASE LAW**
5. **ADMIN DOC**
6. **DOCTRINE**
7. **BOOK**
8. **FOREIGN LAW**

Jayasooriya et al. [84] used Sri Lankan supreme court verdicts as their primary dataset in their experiments of NER and summarisation. The dataset is in English but they had not

[26] Available at https://indiankanoon.org/.

[27] Available at http://www.rechtsprechung-im-internet.de.

provided the number of training and test instances they had used. The processed dataset has not been made available online. The named entities they used in their paper are listed below. However, they do not provide detailed explanations of each named entity.

1. **case titles**
2. **acts**
3. **legal citations**
4. **counsel**
5. **judges**
6. **argued dates**
7. **verdict dates**
8. **judgments**

Shi et al. [20] collected 100 random Chinese judgment documents from available online sources, which are first instance judgments whose civil cause of action is "motor vehicle traffic accident liability dispute". They extract the fact description part of each case and use it for their dataset. Their training, dev and test sets were divided into a ratio of 70:20:10. The named entities they used are listed below. They have not released their dataset.

1. **N**—Natural Person
2. **Np**—Non-natural person
3. **M**—Motor Vehicle
4. **Nm**—Non-motor Vehicles
5. **S**—Insurance
6. **D**—Responsibility
7. **P**—Personal Injury
8. **A**—Article damage
9. **O**—Offence

de Almeida et al. [85] used United States supreme court cases to extract the legal parties involved. They restricted the named entities to only extract the legal parties. And they decided to split their data into 0.8, 0.1 and 0.1 for training, optimisation and testing. They have released the dataset online.[28]

1. **P**—Person
2. **O**—Organisation

Samarawickrama et al. [86] used United States supreme court cases to extract the legal parties involved. They restricted the named entities to only extract the legal parties. They have used 100 test instances in their data and have not released the dataset online.

1. **P**—Person
2. **O**—Organisation

[86] used United States supreme court cases to extract the legal parties involved. They restricted the named entities to only extract the legal parties. They have used 100 test instances in their data and have not released the dataset online.

1. **P**—Person
2. **O**—Organisation

Solihin and Budi [87] introduced a dataset of 150 court decision documents taken from Supreme Court Indonesia.[29] They have not released their dataset online. Following is the list of named entities they used.

28 Available at https://osf.io/mvw7a/?iew_only=.

29 Available at https://putusan.mahkamahagung.go.id/pengadilan.

1. **Nomor Putusan**—Decision Number
2. **Nama terdakwa**—Defendant's name
3. **Tindak pidana**—Criminal action
4. **Melanggar KUHP**—article violation
5. **Tuntutan hukuman**—Penalties
6. **Putusan hukuman**—punishment
7. **Tanggal putusan**—Date of the decision
8. **Hakim Ketua**—presiding judge
9. **Hakim anggota**—judge
10. **Panitera**—Registrar
11. **Penuntut umum**—Prosecutor

Chen et al. [88] used a Chinese dataset of 150,000 rows of data from judicial case texts provided by a middle court in Shandong Province. They used an 8:2 ratio for training and test splits. The authors have not released the dataset online. The following are the named entities they used in their study. They have not provided details about the tagging of the following entities.

1. **Person**
2. **Location**
3. **Organisation**

Sharafat et al. [89] created a NER dataset using 250 civil judgement texts in Lahore High Court. They used k-fold cross-validation for their validation method with k=10. They have not made their dataset freely available online. Following are the 10 types of named entities they used in their study.

1. **CaseNo**—A unique number assigned to each judgment for its identification
2. **Date**—Date of legal judgment.
3. **Loc**—Name of place mentioned in judgment.
4. **Money**—Amount involved in legal judgment.
5. **Org**—Name of an institute or a company.
6. **Per**—Name of a person.
7. **Ref**—Reference to law, act or book.
8. **RefCase**—A reference to a solved case.
9. **RefCourt**—Name of a court that appeared as a reference to a case.
10. **Resp**—Respondent in the case

Legislation and government datasets

Duarte et al. [69] provide a dataset consisting of Portuguese Consumer Law downloaded from Republic Diary,[30] which provides regular rules for the general public in Portuguese. They annotated 36,711 spans on named entities having splits 80%, 10%, and 10% for training, validation and test sets. They have not publicly released the dataset. The five named entity types they used are described below.

1. **LREF**—Legal document references
2. **TREF**—Textual references
3. **NE_ADM**—Names of administrative bodies
4. **TIME_DURATION**—duration
5. **TIME_DATE_REL_TEXT**—Relative dates

LegalNERo [80] consists of files extracted from the Romanian part of the MARCELL[31] corpus. They contain national legislation gathered by crawling from the public Romanian legislative portal.[32] For the LegalNERo corpus, they have selected 370 documents of similar

[30] Available at https://dre.pt/.

[31] Available at https://marcell-project.eu/.

[32] Available at https://legislatie.just.ro/.

size published between 2020–2021. The dataset has 80%, 10% and 10% splits for training, development and testing. The authors have made the data available online.[33] Following are the named entities available.

1. **Person**—Person
2. **Location**—Location
3. **Organization**—Organisation
4. **Time** -Time
5. **Legal Ref**—Legal references

UlyssesNER-Br [92] dataset consists of bills and legislative consultations from Brazilian Chamber of Deputies[34] in Brazilian Portuguese. Training and test splits of the data are 75% and 25%, respectively. The authors have released their data online.[35] Following are the high-level named entities they introduced in their dataset.

1. **DATA**—Date
2. **EVENTO**—Event
3. **FUNDlei**—Legal norm
4. **FUNDapelido**—Legal norm nickname
5. **FUNDprojetodelei**—Bill
6. **FUNDsolicitacaotrabalho**—Legislative consultation
7. **LOCALconcreto**—Concrete place
8. **LOCALvirtual**—Virtual place
9. **ORGpartido**—Political party
10. **ORGgovernamental**—Governmental organisation
11. **ORGnaogovernamental**—Non-governmental organisation
12. **PESSOAindividual**—Individual
13. **PESSOAgrupoind**—Group of individuals
14. **PESSOAcargo**—Occupation
15. **PESSOAgrupocargo**—Group of occupations
16. **PRODUTOsistema**—System product
17. **PRODUTOprograma**—a Program product
18. **PRODUTOoutros**—Others products

Oliveira et al. [94] created a Brazilian Portuguese NER dataset using the Official Gazette of the Federal District Brazil (Diário Ofcial do Distrito Federal [DODF]). DODF includes information about retirements, procurement, decrees, and other matters, organized by the government agency. They annotated the dataset with 11 named entities having 783 training acts, 379 validation acts, and 380 testing acts. They have not released their dataset online. Following are the list of named entities in their dataset.

1. **contract_number**—Contract identification number
2. **GDF_process** —Process number before the Federal District government (GDF)
3. **contractual_parties**—Combination of contracting body, contracted entity, and convening entities
4. **contract_object**—Object to which the contract refers
5. **contract_date**—Contract signature date
6. **contract_value** —Estimated contract final value
7. **contract_duration**—Contract term of validity
8. **budget_unit**—Contract budget union number
9. **work_program**—Contract work program number
10. **nature_of_expenditure**—Contract nature of expenses number
11. **commitment_note**—Contract commitment note

33 Available at https://lod-cloud.net/dataset/racai-legalnero.

34 Available at https://www.camara.leg.br/buscaProposicoesWeb/pesquisaSimplificada.

35 Available at https://github.com/ulysses-camara/ulysses-ner-br.

Contracts and legal agreements

E-NER [72] dataset provides 52 cases from Electronic Data Gathering, Analysis, and Retrieval system (EDGAR),[36] data released by US securities and exchange commission. The dataset consists of 11,696 sentences, and the authors used the k-fold cross-validation technique to evaluate performance with k=5. The dataset is publicly available.[37]

1. **Person**—Person
2. **Location**—Location
3. **Business**—Business organisation
4. **Government**—Government organisation
5. **Court**—Court
6. **Legislation/Act**—Any legislation or mention on Legal Acts
7. **Miscellaneous**—Miscellaneous entities

Chalkidis et al. [90] compiled an annotated English contracts dataset for contract element extraction. The dataset consists of 993 contracts (893 training, 100 test) annotated with clause headings and 2461 contracts (2111 training, 350 test) with gold annotations for the other 11 types of contract elements described below. They have made their dataset publicly available.[38]

1. **Contract Title**—The title of the contract, which usually indicates the type of the contract (e.g. services, employment, loan) and often also the version of the contract (e.g. 'second amendment').
2. **Contracting Parties**—The parties engaged in the contract.
3. **Start Date**—Starting date of the contract.
4. **Effective Date**—Effective date of the contract.
5. **Termination Date**—Termination date of the contract.
6. **Contract Period**—The number of working or calendar days the contract will be effective.
7. **Contract Value**—The price of the agreed transaction (e.g. salary, lump sum)
8. **Governing Law**—The country or state whose laws apply.
9. **Jurisdiction**—The courts responsible to resolve disputes
10. **Legislation Refs**—The laws the contract depends on
11. **Clause Headings**—The heading text which contains the contract title and some important dates.

Aejas et al. [95] introduced a legal contracts NER dataset using data in Electronic Data Gathering, Analysis, and Retrieval (EDGAR) system in United States. They collected 250 contracts in total. Authors have not provided the training and test split ratios. They made their dataset available online.[39] Following are the named entity tags they annotated in their dataset.

[36] Available at https://www.sec.gov/os/accessing-edgar-data.

[37] Available at https://github.com/terenceau2/E-NER-Dataset.

[38] Available at http://nlp.cs.aueb.gr/software.html.

[39] Available at https://drive.google.com/file/d/1cJmqifHzuR8G1bIfeSvncP98exqBnBQM/view.

1. **Parties**—The individuals or organisations who are involved in the agreement are commonly known as parties.
2. **Title**—The contract's type, such as a supply agreement or a co-branding agreement, is indicated by the title.
3. **EffectiveDate**—The effective date is an essential component of a contract as it determines the exact date from which the agreement becomes enforceable.
4. **NotificationPeriod**—Notification Period refers to the amount of time, usually in days or months, that is required for a party to provide written notification to the other party before terminating the contract.
5. **TermLength**—The term length refers to the duration of the contract period, which may be specified in the contract, or replaced by the termination date.
6. **GoverningLaw**—The governing law specifies which country or state's law will be followed in case of any legal disputes arising from the contract.

Specialized/domain-specific datasets

Orasmaa et al. [74] released a dataset containing Estonian Parish Court records from the 1820s to the beginning of the 20th century. The majority of texts originate from the period of 1860–1890. The original documents are handwritten, and volunteers digitise them manually. The authors mention that the dataset is not completely error free, but the dataset has a huge value for NLP researchers as well as historians. The authors used 21,546 sentences in splits of 16,040 for training, 2336 for the development set, and 3170 for the test set. The dataset is publicly available online at their GitHub repository.[40]

1. **PER**—Full name of a person
2. **LOC-ORG**—A place of a group of people connected to a location
3. **LOC**—Names of landscape objects
4. **ORG**—Parish courts and higher courts
5. **MISC**—Unknown named entities

Andrew [81] used a French corpus taken from the so-called "Luxembourg" corpus. The data contains information about people and companies who are investing money or property in the state of Luxembourg. Their training data contained 35 text files and the test set is a collection of 21 text documents. The data is not publicly available. The named entities are described below.

1. **PERSONNE**—Name of the person
2. **NOM**— First name of the person
3. **ADDRESS**—Organisation address
4. **SOCIETE_PRINCIPALE**—Main company in the transaction
5. **SOCIETE_SECONDAIRE**—Secondary companies in the transaction
6. **ROLE**—Role of the person
7. **FONCTION**—Function or position
8. **TYPE_SOCIETE**—Type of companies

Schraagen et al. [91] created an annotated dataset of 250 criminal complaint reports submitted to the official website of the Dutch police in the domain of internet fraud. They did not use a training set in their methodology; hence, the whole dataset is for testing. They have not released the data online. The following is the list of named entities they studied, but they have not mentioned a detailed explanation for each tag.

[40] Available at: https://github.com/soras/vk_ner_lrec_2022.

1. **location**
2. **person**
3. **organisation**
4. **event**
5. **product**
6. **miscellaneous**

Kwak et al. [93] created a dataset consisting of 26 entities using 45 wills from Ancestry, which contains documents in the public domain. The wills are from two US states (Tennessee and Idaho) and their execution dates range from 1978 to 2001. Their training, dev and test sets contain 30, 5 and 5 wills, and they name the 5 wills from Idaho as out-of-domain sets. They have released their data online.[41] Following list shows their named entities and short descriptions of them.

1. **Testator**—A person who makes a will.
2. **Trigger**—Event trigger words
3. **Condition**—A condition under which an event occurs
4. **Beneficiary**—A person or entity that receives something from a will
5. **Will**—A legal document containing a person's wishes regarding the disposal of one's asset after death
6. **Asset**—Any money, personal property, or real estate owned by a testator
7. **Executor**—A person who executes a will
8. **Witness**—A person witnessing a will
9. **Time**—Any expression denoting a particular point in time
10. **Duty**—Any duty directed by a testator to fiduciaries
11. **State**—Any US state names
12. **Date**—Any dates
13. **County**—Any US county names
14. **Right**—Any rights authorised by a testator to fiduciaries
15. **Expense**—Any expenses
16. **Debt**—Any debts
17. **Bond**—Any bonds
18. **Codicil**—A testamentary or supplementary document that modifies or revokes a will or part of a will
19. **Trustee**—A person who manages a trust
20. **Non-beneficiary**—A person who is excluded from being beneficiary
21. **Tax**—Any taxes
22. **Affidavit**—A legal statement sworn and signed by a testator and witnesses to confirm the validity of a will
23. **Notary public**—A person who is authorised by state government to witness the signing of important documents and administer oaths
24. **Trust**—A fiduciary arrangement that allows a trustee to hold assets on behalf of a beneficiary
25. **Guardian**—A person who has a legal right and responsibility of taking care of someone who cannot takes care of themselves
26. **Conservator**—A person who handles the financial and personal affairs who cannot handles such affairs by themselves

Appendix B: Overview of legal RE datasets

In the following list, we describe the legal RE datasets, elaborating on the relationships used.

Court judgments and case law RE datasets

Chen et al. [160] dataset consists of 1750 fact descriptions of the drug-related criminal judgment documents downloaded from China Judgments Online. From 12 drug-related crime types, they have focused on three types, i.e. drug trafficking, illegal possession of drugs and

[41] Available at https://github.com/ml4ai/ie4wills.

providing venues for drug users. They split the dataset by a ratio of 4:1 to obtain the training set and the test. They have not made their dataset publicly available.

1. **traffic_in**—Suspect deals in drugs.
2. **sell_drug_to**—Conducts a drug trade with someone else.
3. **possess**—Suspect holds a certain amount of narcotic drugs.
4. **provide_shelter_for**—Suspect provides shelter for others to ingest or inject drugs.

Kurant [162] released a RE dataset based on 150 Polish judgements. The author filtered the sentences which contain cause-and-effect relations, and due to the small number of sentences in the case-effect-relation dataset, the author used 10-fold cross-validation in their validation process. The author has not released the dataset.

1. **cause-effect-relation**—relationship between events e1 and e2, such that the occurrence of event e1 results in the occurrence of event e2.

Legislation and regulatory RE datasets

Zhong et al. [161] introduce a dataset with five relationships in construction procedural constraints in Chinese regulations. Training and test splits were 70% and 30% of data. They have not made their data freely available. They use a convention of categories A and B, where Category-A is the relationship between two specific construction activities, and Category-B is the relationship between a construction activity and a certain construction state. Following are the relationships they annotated in their dataset.

1. **Pattern A1**— belonging to Category-A and mutually-exclusive relationships; Meet
2. **Pattern A2**—belonging to Category-A and mutually-exclusive relationships; Before
3. **Pattern A3**—belonging to Category-A and mutually-exclusive relationships; Before
4. **Pattern A4**—belonging to Category-A and mutually-exclusive relationships; Before
5. **Pattern B**—Category-B

Contracts, companies RE datasets

Andrew [81] introduced a French dataset for relationship extraction, which contains information about people and companies who are investing money or property in the state of Luxembourg. Training and test data were created using 35 and 21 documents, respectively. They follow a *"distant supervision"* approach [173] where all the sentences containing two specific named entities are considered as carrying a relationship (We described their named entities in the previous section). Therefore, the author has not used human annotations for the relationships. The data is not publicly available.

1. **PERSONNE_FONCTION**— the relationship between the class "PERSONNE" and the class "FONCTION"
2. **PERSONNE_ROLE**—the relationship between the class "PERSONNE" and the class "ROLE"

3. **SOCIETE_ROLE**—the relationship between the class "SOCIETE" and the class "ROLE"
4. **SOCIETE_TYPE**—the relationship between the class "SOCIETE" and the class "TYPE_SOCIETE"

Specialized/domain-specific RE datasets

Schraagen and Bex [163] apply relationship extraction to user-generated police reports in Dutch. The data has been collected through a web form in the domain of internet fraud. They have used 10-fold cross-validation for their validation process, and they have not released their dataset freely available.

1. **residency**—the place they live or lived.

Kwak et al. [93] created a dataset consists of 18 relationship types using 45 wills from United States. The wills are from two US states (Tennessee and Idaho) and their execution dates range from 1978 to 2001. Their training, dev and test sets contain 30, 5, 5 wills and they name the 5 wills from Idaho as out-of-domain set. They have released their data online.[42] The relationship types are listed below.

1. **Coreference resolution**—The relation between entities referring to the same real-world entity
2. **Beneficiary-Asset**—The relation between a beneficiary and asset
3. **Testator-Asset**—The relation between a testator and asset
4. **Testator-Beneficiary**—The relation between a testator and a beneficiary
5. **Testator-Will**—The relation between a testator and a will
6. **Testator-Executor**—The relation between a testator and an executor
7. **Inclusion**—The relation between an entity and another entity in which the former one includes the latter one
8. **Competence**—The relation between a testator and a condition that shows the competence of the testator
9. **Parent-Child**—The relation between a parent and a child
10. **Witness-Testator**—The relation between a witness and a testator
11. **Witness-Will**—The relation between a witness and a will
12. **Spouse-Spouse**—Spousal relations
13. **Beneficiary-Executor**—The relation between a beneficiary and an executor
14. **Beneficiary-Will**—The relation between a beneficiary and a will
15. **Testator-Codicil**—The relation between a testator and a codicil
16. **Testator-Trustee**—The relation between a testator and a trustee
17. **Testator-Guardian**—The relation between a testator and a guardian
18. **Testator-Conservator**—The relation between a testator and a conservator

Appendix C: Overview of legal ED datasets

In the following list, we describe the legal ED datasets, elaborating on different event types and event arguments they present.

[42] Available at https://github.com/ml4ai/ie4wills.

Court judgments and case law event detection datasets

LEVEN [188] features 108 event types, over 150k event mentions, using over 8k documents in criminal law. They mostly identify the events based on the case law provided by the Chinese government website.[43] Most of the event types are covered in the case documents, but they have enriched the less-frequent event types by using law handbooks. Their dataset consists of 41,238, 9788 and 12,590 sentences in training, validation and test sets. They have released their dataset online[44]. [45]

Li et al. [189] created an event extraction dataset consisting of 5 main event types related to larceny law. They also introduced 6 event arguments and 5 event argument roles. The main source of data is the CAIL 2018 [236] dataset.[46] Out of 3000 larceny related cases, they selected 6538 sentences which contained the event types they defined. They have not provided their dataset online.

Following are the event types they used in their dataset.

1. **Steal**—Take something from a person or an organisation secretly
2. **Draw**—Withdraw or transfer money from a stolen bank account/card
3. **Spend Money**—Spend stolen money
4. **Handle Things**—Handle stolen goods, such as selling, discarding, etc.
5. **Surrender**—The criminal surrender himself/herself

The event arguments they used are listed below.

1. **Time**—Time of the event
2. **Person**—Participants of the event
3. **Location**—Where the event occurred
4. **Amount**—Amount involved in the event
5. **Thing**—Things involved in the event
6. **Number**—Number of the things involved in the event

Following is the list of argument roles introduced in their paper.

1. **Steal**—Steel Time, Crime Suspect, Victim, Steal Location, Steal Thing, Steal Number, Steal Amount
2. **Draw**—Draw Time, Depositor, Draw Location, Draw Amount
3. **Spend Money**—Spend Time, Spender, Spend Location, Spend Amount
4. **Handle Things**—Handle Time, Handler, Handled Thing, Handle Amount
5. **Surrender**—Crime Suspect

Shen et al. [190] manually labelled a Chinese legal event extraction dataset from the Chinese legal document corpus created using the Chinese government website. The dataset contains 2380 instances with 11 pre-defined event types, 26 pre-defined event-argument roles, and 17 pre-defined argument-argument roles. They selected 70% and 30% of the instances as the training and test data splits, respectively. The annotated dataset has not been released online. Following are the event types defined in their study.

[43] Available at https://wenshu.court.gov.cn/.

[44] Available at https://github.com/thunlp/LEVEN.

[45] Please refer to the original paper for the complete set of event types.

[46] Available at http://cail.cipsc.org.cn/.

1. **RECKLESS DRIVING**—refers to driving a car on the road: chase racing, drunk driving, overloading, speeding, and other actions that endanger public safety.
2. **TRAFFIC OFFENSE**—refers to the criminal act of violating road traffic management laws and regulations, causing serious traffic accidents, causing serious injury or death, or causing heavy losses to public and private property being prosecuted for criminal responsibility according to law.
3. **LARCENY**—refers to the act of illegal possession, the theft of public and private property objects or multiple theft, household theft, theft with a weapon, and pickpocketing of public and private property.
4. **FRAUD**—refers to the act of deceiving large amounts of public and private property for the purpose of illegal possession, using fictitious facts or concealing the truth.
5. **ROBBERY**—is an act of illegal possession, using violence, coercion or other methods against the owner or custodian of property to steal public and private property forcefully.
6. **INTENTION INJURY**—refers to the crime convicted of intentionally illegally harming the health of others.
7. **INTENTION KILLING**—refers to the act of deliberately depriving others of their lives.
8. **ARREST**—refers to a case where a public security organ, a people's procuratorate, or a people's court deprives a criminal suspect or defendant of an act that impedes criminal proceedings, evades investigation, prosecution, trial, or social danger, and deprives him of his personal freedom under the law and is detained. Kind of coercive measures.
9. **DETENTION**—refers to a compulsory state that restricts the personal freedom of a criminal suspect or defendant detained or arrested within a certain period of time.
10. **BAIL**—release refers to the criminal defendant detained in the judiciary providing security and granting release.
11. **TRIAL**—means hearing a case and giving a judgment. It is an important part of law enforcement power.

The following list refers to all entity argument types involved in the hierarchical event.

1. **PARTY**—is a person who enters a lawsuit because of a dispute over the rights and interests in the substantive law or has a direct relationship with a specific legal fact and is bound by a court decision.
2. **ITEM**—refers to the items involved in the crime.
3. **TIME**—refers to the time of the event.
4. **LOCATION**—refers to the location where the crime occurred.
5. **ORGANIZATION**—refers to law enforcement agencies, procuratorates, and courts.
6. **TERM**—is an important basis for conviction and sentencing.
7. **CRIMINAL CHARGE**—is the name or title of the crime, and it is a high-level summary of the essential characteristics or main characteristics of the crime.
8. **BEHAVIOR**—refers to the appearance of activities that are controlled by ideas. This article refers to the actions taken by the parties before or after the crime.
9. **PHYSICAL STATE**—refers to the disability and mental state of the party.
10. **BEHAVIORAL STATE**—refers to the state of the party's behaviour, such as epilepsy and drunkenness.
11. **ITEM ATTRIBUTE**—is an abstract characterisation of items, such as value, length, and diameter. The article

attributes mainly relate to the value of the article.

12. **PARTY ATTRIBUTE**—refers to the attribute information of the party's ethnicity and age.
13. **CRIME ATTRIBUTE**—describes the social impact brought by criminal behaviour, whether the defendant's criminal methods are cruel, and the reasons for the occurrence of criminal incidents.
14. **PENALTY/ENFORCE ATTRIBUTE**—is a more detailed description of the punishment of the defendant's criminal facts.

Li et al. [191] created event types for event extraction in divorce cases in China. They crafted 13 event types and annotated 3100 cases for event extraction. They have not released their data online.

Following are the 13 types of focus events for divorce cases.

1. **Know**—(KnowTime, Participant1, Participant2)
2. **BeInLove**—(BeInLoveTime, Participant1, Participant2)
3. **Marry**—(MarryTime, Participant1, Participant2)
4. **Remarry**—(Participant)
5. **Bear**—(Time, Gender, Name, Age)
6. **FamilyConflict**—(Participant1, Participant2)
7. **DomesticViolence**—(Time, Perpetrators, Victim)
8. **BadHabit**—(Participant)
9. **Derailed**—(Time, Derailer, DerailedTarget)
10. **Separation**—(BeginTime, EndTime, Duration)
11. **DivorceLawsuit**—(SueTime, Initiator, Court, JudgeTime, JudgeDocument, Result)
12. **Wealth**—(Value, IsCommon, IsPersonal, Whose)
13. **Debt**—(Value, Debtor, Creditor)

Araujo et al. [194] utilised a corpus of 200 Brazilian legal documents, composed of 39,895 sentences, obtained from RS State Superior Court (in Portuguese, Tribunal Superior do Rio Grande do Sul—TJRS). The event types used in their experiment are listed below. However, the authors have not mentioned the training to test ratio neither released the data.

1. **Denúncia**—Formal charges
2. **Absolvição**—Acquittal
3. **Condenação**—Conviction
4. **Interrogatório**—Questioning

Kwak et al. [93] published a dataset consisting of 20 event types using 45 wills from the United States. The wills are from two US states (Tennessee and Idaho) and their execution dates range from 1978 to 2001. Their training, dev and test sets contain 30, 5, and 5 wills, and they name the 5 wills from Idaho as out-of-domain sets. They have released their data online.[47] The event types are listed below.

1. **Bequest**—An event in which a testator bequeath asset to a beneficiary
2. **Sign will**—An event in which a testator or a witness signs a will
3. **Direction**—An event in which a testator gives direction to a fiduciary
4. **Will creation**—An event in which a testator creates a will
5. **Attestation**—An event in which a witness attests the validity of a will
6. **Excuse**—An event in which a testator excuses a fiduciary from a duty

[47] Available at https://github.com/ml4ai/ie4wills.

7. **Authorization**—An event in which a testator authorises a fiduciary to a right
8. **Nomination**—An event in which a testator nominates a fiduciary
9. **Death**—An event in which any entity
10. **Revocation**—An event in which a testator revokes a will or a codicil
11. **Probate**—An event in which a will or any part of the will is probated
12. **Codicil**—An event in which a codicil is made
13. **Disqualification**—An event in which a beneficiary or a fiduciary is disqualified
14. **Removal**—An event in which a beneficiary is removed from the will
15. **Give**—An event in which a testator gives a compensation to a fiduciary
16. **Non probate instrument creation**—An event in which a non probate instrument is created
17. **Renunciation**—An event in which a fiduciary renounces
18. **Notarization**—an event in which an affidavit is notarised by a notary public
19. **Birth**—An event in which a beneficiary is born
20. **Residual**—An event in which asset becomes residuary estate

Contracts, companies and financial event detection datasets

Paulino-Passos et al. [193] consists of annotations for logic programming applications on contract documents. They have gathered named entities and contract events using the United States Caselaw database.[48] The data consists of 207 unique documents with a total of 3934 sentences. Their paper does not explore any experiments on event detection. Therefore, the dataset has no training and test splits. They have made their dataset available online.[49] Following are the event types they annotated in their work. However, they have not provided descriptions of them.

1. **Purchase-contract**
2. **Other-contract**
3. **Negotiation**
4. **Agreement**
5. **Execution**
6. **Rescission**
7. **Expiration**

Specialized/domain-specific event detection datasets

Filtz et al. [192] created a corpus[50] consisting of 30 decisions from the European Court of Human Rights (ECHR). The primary focus of this dataset is to identify events to automate court decision timeline generation such as events, processes, temporal information, and the parties involved. Their argument annotation is primarily based on What-When-Who questions. The dataset has been divided into a training set with 24 documents and test and validation sets with 3 documents each. They have made their data online.[51]
Their annotation consists of the following event types.

[48] Available at https://case.law.

[49] Available at https://zenodo.org/records/8098312.

[50] Available at https://zenodo.org/records/4032617.

[51] Available at https://github.com/mnavasloro/EventsMatter.

1. **circumstance**—meaning that the event corresponds to the facts under judgment.
2. **procedure**—wherein the event belongs to the procedural dimension of the case.

Their event argument detection is based on What-When-Who questions, but they have not provided details of exact event arguments.
Kwak et al. [93] published a dataset consisting of 20 event types using 45 wills from the United States. The wills are from two US states (Tennessee and Idaho) and their execution dates range from 1978 to 2001. Their training, dev and test sets contain 30, 5, and 5 wills, and they name the 5 wills from Idaho as out-of-domain sets. They have released their data online.[52] The event types are listed below.

1. **Bequest**—An event in which a testator bequeath asset to a beneficiary
2. **Sign will**—An event in which a testator or a witness signs a will
3. **Direction**—An event in which a testator gives direction to a fiduciary
4. **Will creation**—An event in which a testator creates a will
5. **Attestation**—An event in which a witness attests the validity of a will
6. **Excuse**—An event in which a testator excuses a fiduciary from a duty
7. **Authorization**—An event in which a testator authorises a fiduciary to a right
8. **Nomination**—An event in which a testator nominates a fiduciary
9. **Death**—An event in which any entity
10. **Revocation**—An event in which a testator revokes a will or a codicil
11. **Probate**—An event in which a will or any part of the will is probated
12. **Codicil**—An event in which a codicil is made
13. **Disqualification**—An event in which a beneficiary or a fiduciary is disqualified
14. **Removal**—An event in which a beneficiary is removed from the will
15. **Give**—An event in which a testator gives a compensation to a fiduciary
16. **Non probate instrument creation**—An event in which a non probate instrument is created
17. **Renunciation**—An event in which a fiduciary renounces
18. **Notarization**—an event in which an affidavit is notarised by a notary public
19. **Birth**—An event in which a beneficiary is born
20. **Residual**—An event in which asset becomes residuary estate

Xian et al. [195] introduced a document level event extraction dataset using existing Chinese case retrieval dataset [237] and by collecting judgements from an online available source.[53] The annotated data for each event type was randomly divided into training, test, and development sets in an approximate ratio of 8:1:1. Their dataset is available on request. However, they made their dataset schema online available.[54] Following are the top level event types they experimented with.

1. **Disrupting market economic order**
2. **Misconduct**
3. **Infringing property**
4. **Endangering public safety**
5. **Disturbing public order**
6. **Bribery and corruption**
7. **Violating democratic rights**

52 Available at https://github.com/ml4ai/ie4wills.

53 Available at https://wenshu.court.gov.cn/.

54 Available at https://anonymous.4open.science/r/DLEE-DATA/README.md.

They have not provide the full list of arguments in their paper.

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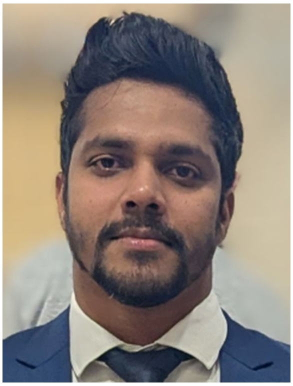

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